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BEGUM ROKEYA UNIVERSITY, RANGPUR

Department of Statistics

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## ASSIGNMENT

*On*

Categorical Data Analysis

**Name :** Md Montajul Islam

**ID :** 12110020

**Registration No. :** 000015883

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1. Consider the lung cancer data and solve the following questions using Python.

Basic EDA Questions i) Summarize the distribution of Age. Is it normally distributed or skewed? ii) What proportion of individuals are smokers vs non-smokers? iii) Which income group has the highest frequency? iv) What percentage of individuals are exposed to high pollution? v) What is the overall prevalence of lung disease?

Association & Crosstab i) Is there an association between Smoking and Lung Disease? ii) Interpret using a contingency table. iii) Does Pollution level affect Lung Disease? iv) Compare lung disease prevalence across Income groups. v) Which factor appears to have the strongest relationship with lung disease?

Statistical Testing i) Perform a Chi-square test for: Smoking vs Lung Disease and Pollution vs Lung Disease. ii) When should you use Fisher's Exact Test instead of Chi-square? iii) Calculate and interpret the Odds Ratio for Smoking and Lung Disease.

Correlation & Interpretation i) Interpret the correlation between: Smoking and Lung Disease, Age and Lung Disease. ii) Why is correlation not sufficient for causal inference?

Logistic Regression i) Fit a logistic regression model: Lung Disease ~ Smoking + Age + Pollution + Income. ii) Which variables are statistically significant predictors? ii) Interpret the odds ratio of Smoking. iii) What happens to the effect of Smoking after adjusting for Pollution?

Advanced Modeling i) Add an interaction term (Smoking x Pollution). ii) Is the interaction significant? iii) Interpret the interaction: Does pollution amplify smoking risk?

Research question: using the lung disease dataset, perform exploratory data analysis and fit a logistic regression model to identify key determinants of lung disease. interpret your finds and discuss public health implications. now give solution using python

```
[10]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from scipy.stats import chi2_contingency, fisher_exact

import statsmodels.api as sm

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    confusion_matrix,
    classification_report,
    roc_curve,
    roc_auc_score
)

# 2. LOAD DATASET
df = pd.read_csv("lung_disease.csv")
```

```

print("\nFIRST 5 ROWS")
print(df.head())

print("\nCOLUMN NAMES")
print(df.columns)

print("\nDATA INFO")
print(df.info())

# 3. BASIC EDA

# AGE DISTRIBUTION

print("\nAGE SUMMARY")
print(df['Age'].describe())

sns.histplot(df['Age'], kde=True, color='skyblue')

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()

# SMOKERS VS NON-SMOKERS

print("\nSMOKING COUNTS")
print(df['Smoking'].value_counts())

sns.countplot(x='Smoking', data=df)

plt.title("Smoking Distribution")

plt.show()

# INCOME GROUP FREQUENCY

print("\nINCOME GROUP COUNTS")

income_counts = df['Income'].value_counts()

print(income_counts)

print("\nMOST COMMON INCOME GROUP")
print(income_counts.idxmax())

# HIGH POLLUTION PERCENTAGE

```

```

high_pollution_percentage = (
    (df['Pollution'] == 'High').mean() * 100
)

print(f"\nHIGH POLLUTION EXPOSURE: {high_pollution_percentage:.2f}%")

# LUNG DISEASE PREVALENCE

lung_prevalence = (
    (df['LungDisease'] == 1).mean() * 100
)

print(f"\nLUNG DISEASE PREVALENCE: {lung_prevalence:.2f}%")

# 4. CROSSTAB & ASSOCIATION

# SMOKING VS LUNG DISEASE

crosstab_smoking = pd.crosstab(
    df['Smoking'],
    df['LungDisease']
)

print("\nCROSSTAB: SMOKING VS LUNG DISEASE")
print(crosstab_smoking)

sns.heatmap(
    crosstab_smoking,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.title("Smoking vs Lung Disease")

plt.show()

# 5. POLLUTION EFFECT

income_disease = pd.crosstab(
    df['Income'],
    df['LungDisease'],
    normalize='index'
) * 100

print("\nLUNG DISEASE BY INCOME GROUP (%)")

```

```

print(income_disease)

income_disease.plot(kind='bar', stacked=True)

plt.title("Lung Disease Across Income Groups")
plt.ylabel("Percentage")

plt.show()

# 6. STATISTICAL TESTING

# CHI-SQUARE TEST

chi2, p, dof, expected = chi2_contingency(crosstab_smoking)

print("\nCHI-SQUARE TEST")

print("Chi-square statistic:", chi2)
print("P-value:", p)

if p < 0.05:
    print("Significant association exists")
else:
    print("No significant association")

# POLLUTION VS LUNG DISEASE

crosstab_pollution = pd.crosstab(
    df['Pollution'],
    df['LungDisease']
)

chi2_p, p_p, dof_p, expected_p = chi2_contingency(
    crosstab_pollution
)

print("\nPOLLUTION VS LUNG DISEASE")

print("Chi-square statistic:", chi2_p)
print("P-value:", p_p)

# FISHER'S EXACT TEST

print("""

```

```

Use Fisher's Exact Test when:
- Sample size is small
- Expected counts are low
- Data is 2x2
"""

if crosstab_smoking.shape == (2, 2):

    oddsratio, fisher_p = fisher_exact(crosstab_smoking)

    print("\nFISHER'S EXACT TEST")

    print("Odds Ratio:", oddsratio)
    print("P-value:", fisher_p)

# ODDS RATIO

if crosstab_smoking.shape == (2, 2):

    a, b, c, d = crosstab_smoking.values.flatten()

    odds_ratio = (a * d) / (b * c)

    print("\nODDS RATIO:", odds_ratio)

    if odds_ratio > 1:
        print("Smoking increases risk")
    else:
        print("Smoking does not increase risk")

# 7. CORRELATION ANALYSIS

# ENCODE CATEGORICAL VARIABLES

encoded_df = df.copy()

mapping = {
    'Yes': 1,
    'No': 0,
    'High': 2,
    'Medium': 1,
    'Low': 0
}

for col in encoded_df.columns:

    if encoded_df[col].dtype == 'object':

```

```

        encoded_df[col] = encoded_df[col].map(mapping)

# CORRELATION MATRIX

corr = encoded_df.corr(numeric_only=True)

print("\nCORRELATION MATRIX")
print(corr)

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Matrix")

plt.show()

# 8. LOGISTIC REGRESSION

# CONVERT TARGET VARIABLE TO NUMERIC

# Convert Yes/No → 1/0

df['LungDisease'] = df['LungDisease'].map({
    'Yes': 1,
    'No': 0
})

# DEFINE TARGET VARIABLE

y = df['LungDisease']

# DEFINE FEATURES

X = df.drop(columns=['LungDisease'])

# CONVERT CATEGORICAL VARIABLES

X = pd.get_dummies(X, drop_first=True)

# CONVERT TO NUMERIC

```



```

X = X.astype(float)

y = y.astype(int)

# ADD CONSTANT

X = sm.add_constant(X)

# CHECK DATA TYPES

print(X.dtypes)

# FIT LOGISTIC REGRESSION

log_model = sm.Logit(y, X).fit()

# SUMMARY

print(log_model.summary())

# ODDS RATIOS

odds_ratios = pd.DataFrame({
    'Variable': log_model.params.index,
    'Odds Ratio': np.exp(log_model.params.values)
})

print("\nODDS RATIOS")
print(odds_ratios)

```

FIRST 5 ROWS

|   | ID | Smoking | Age | Income | Pollution | LungDisease |
|---|----|---------|-----|--------|-----------|-------------|
| 0 | 1  | Yes     | 36  | Low    | High      | Yes         |
| 1 | 2  | No      | 28  | Low    | Low       | No          |
| 2 | 3  | No      | 52  | Medium | High      | No          |
| 3 | 4  | No      | 39  | High   | High      | No          |
| 4 | 5  | Yes     | 32  | Low    | Low       | Yes         |

COLUMN NAMES

```
Index(['ID', 'Smoking', 'Age', 'Income', 'Pollution', 'LungDisease'],
      dtype='object')
```

DATA INFO

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 500 entries, 0 to 499

Data columns (total 6 columns):

| # | Column      | Non-Null Count | Dtype  |
|---|-------------|----------------|--------|
| 0 | ID          | 500 non-null   | int64  |
| 1 | Smoking     | 500 non-null   | object |
| 2 | Age         | 500 non-null   | int64  |
| 3 | Income      | 500 non-null   | object |
| 4 | Pollution   | 500 non-null   | object |
| 5 | LungDisease | 500 non-null   | object |

dtypes: int64(2), object(4)

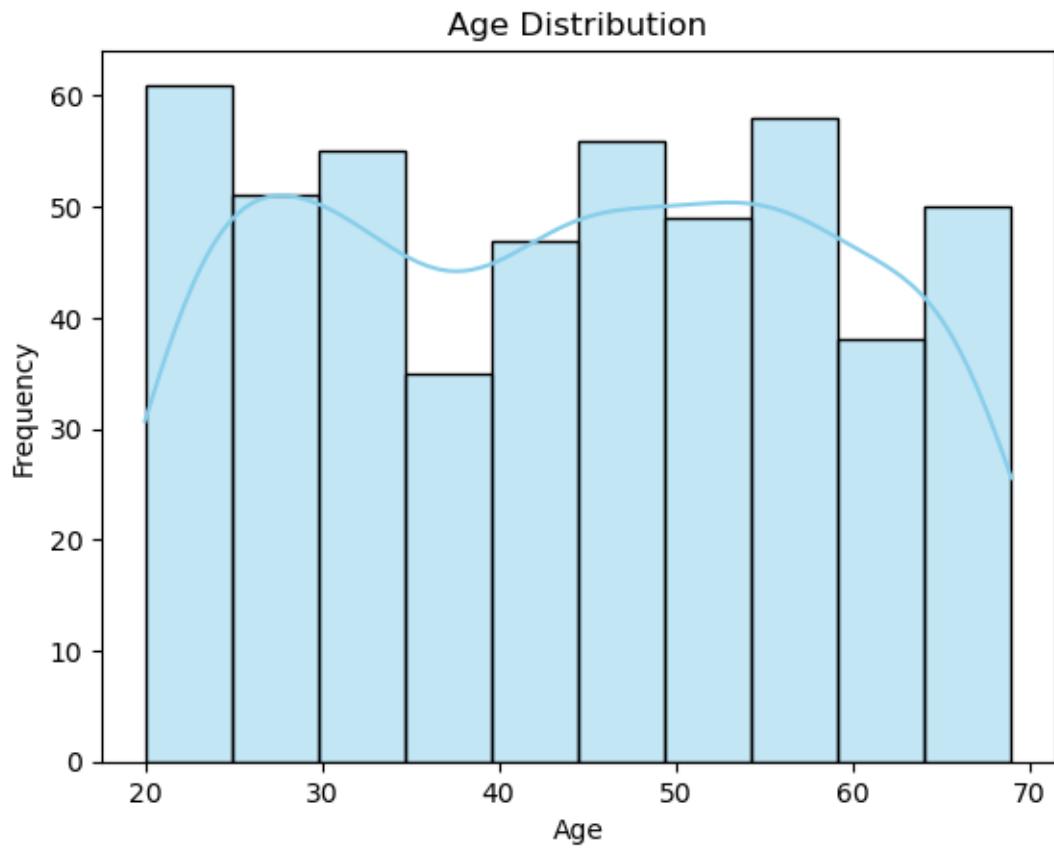
memory usage: 23.6+ KB

None

#### AGE SUMMARY

|       |            |
|-------|------------|
| count | 500.000000 |
| mean  | 43.904000  |
| std   | 14.604841  |
| min   | 20.000000  |
| 25%   | 30.750000  |
| 50%   | 45.000000  |
| 75%   | 56.000000  |
| max   | 69.000000  |

Name: Age, dtype: float64



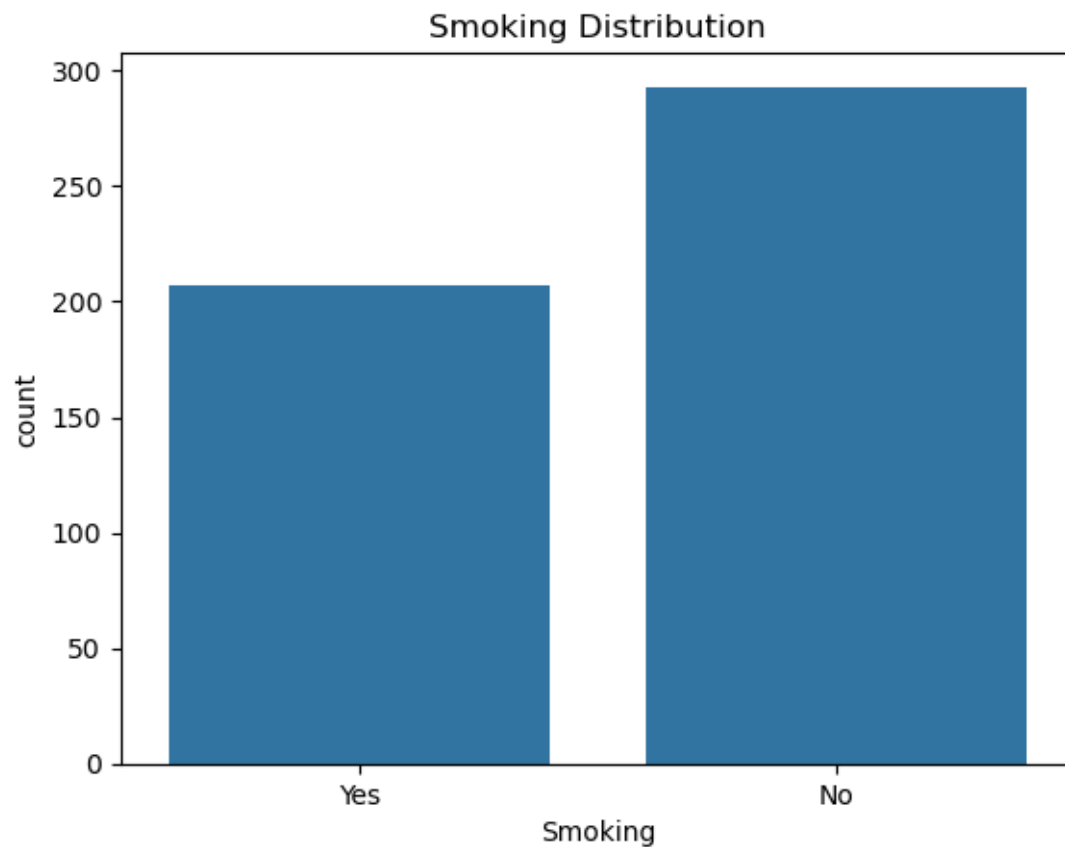
#### SMOKING COUNTS

Smoking

No 293

Yes 207

Name: count, dtype: int64



#### INCOME GROUP COUNTS

Income

Low 204

Medium 197

High 99

Name: count, dtype: int64

#### MOST COMMON INCOME GROUP

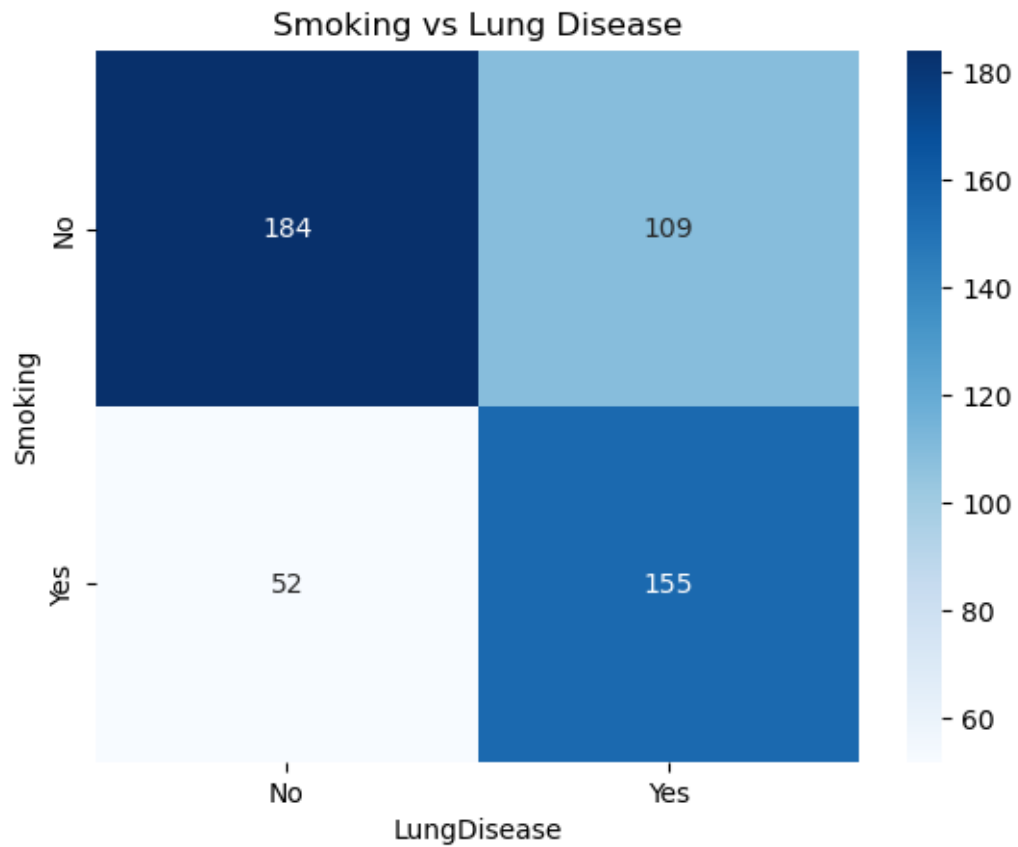
Low

HIGH POLLUTION EXPOSURE: 70.40%

LUNG DISEASE PREVALENCE: 0.00%

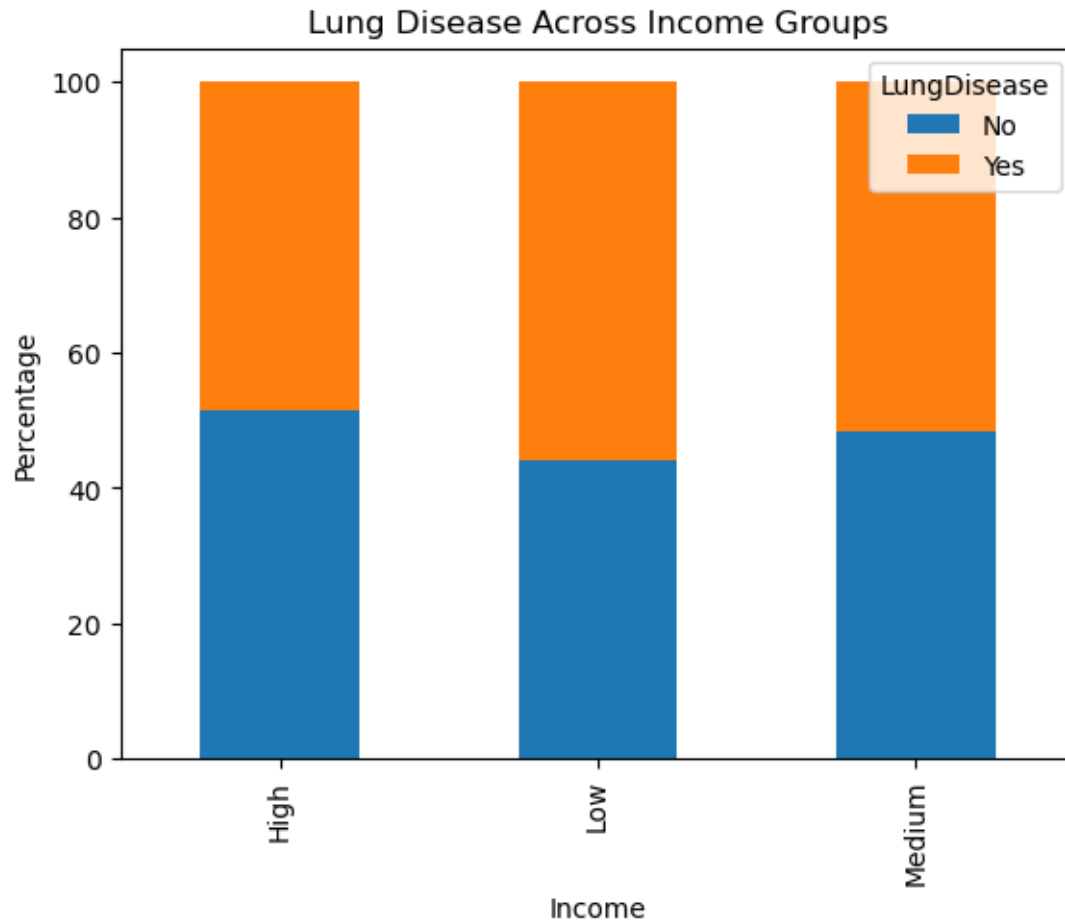
#### CROSSTAB: SMOKING VS LUNG DISEASE

| LungDisease | No  | Yes |
|-------------|-----|-----|
| Smoking     |     |     |
| No          | 184 | 109 |
| Yes         | 52  | 155 |



LUNG DISEASE BY INCOME GROUP (%)

|             |           |           |
|-------------|-----------|-----------|
| LungDisease | No        | Yes       |
| Income      |           |           |
| High        | 51.515152 | 48.484848 |
| Low         | 44.117647 | 55.882353 |
| Medium      | 48.223350 | 51.776650 |



#### CHI-SQUARE TEST

Chi-square statistic: 67.5942649754021

P-value: 2.0085094601569646e-16

Significant association exists

#### POLLUTION VS LUNG DISEASE

Chi-square statistic: 33.84256825473482

P-value: 5.975692575712951e-09

Use Fisher's Exact Test when:

- Sample size is small
- Expected counts are low
- Data is 2x2

#### FISHER'S EXACT TEST

Odds Ratio: 5.031757233592096

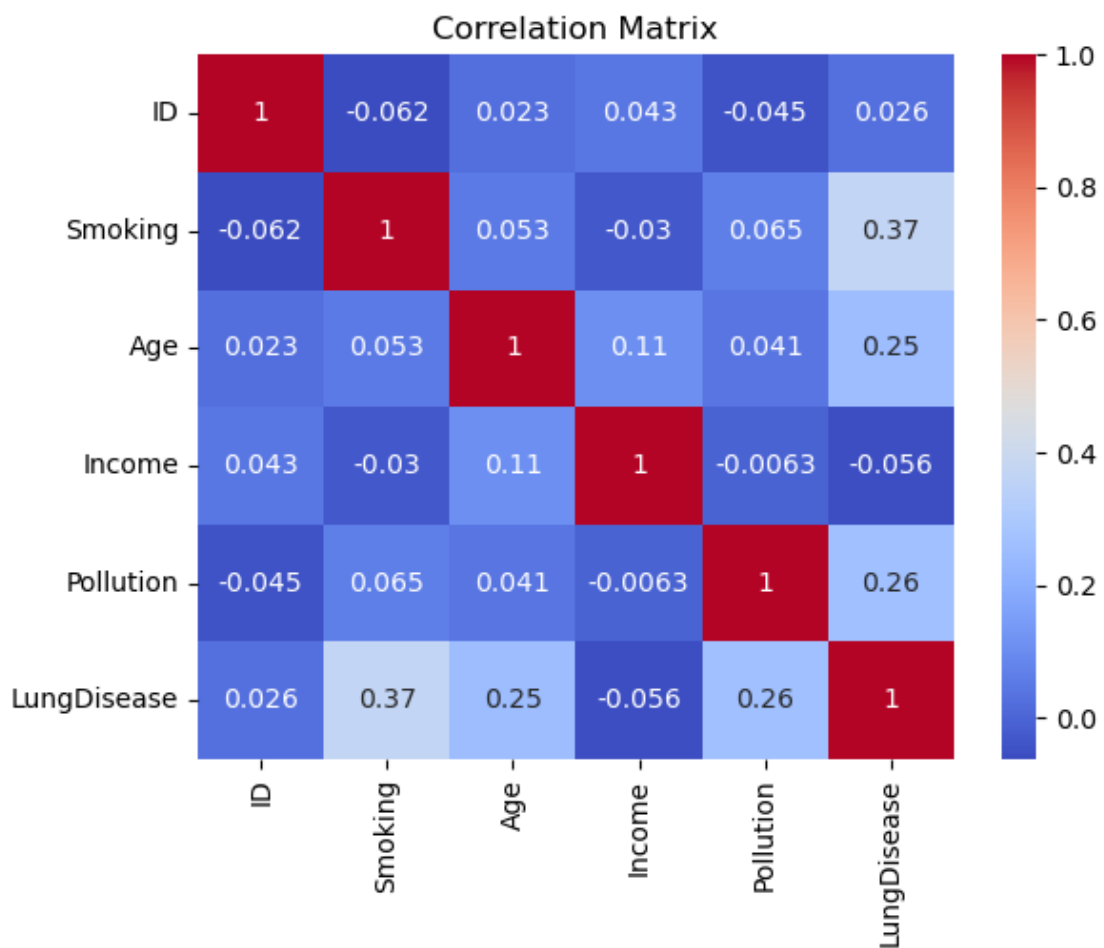
P-value: 5.128610707567623e-17

ODDS RATIO: 5.031757233592096

Smoking increases risk

#### CORRELATION MATRIX

|             | ID        | Smoking   | Age      | Income    | Pollution | LungDisease |
|-------------|-----------|-----------|----------|-----------|-----------|-------------|
| ID          | 1.000000  | -0.062467 | 0.023478 | 0.042987  | -0.044530 | 0.025980    |
| Smoking     | -0.062467 | 1.000000  | 0.052564 | -0.029956 | 0.064685  | 0.371747    |
| Age         | 0.023478  | 0.052564  | 1.000000 | 0.107329  | 0.040776  | 0.249694    |
| Income      | 0.042987  | -0.029956 | 0.107329 | 1.000000  | -0.006312 | -0.056439   |
| Pollution   | -0.044530 | 0.064685  | 0.040776 | -0.006312 | 1.000000  | 0.264552    |
| LungDisease | 0.025980  | 0.371747  | 0.249694 | -0.056439 | 0.264552  | 1.000000    |



|             |         |
|-------------|---------|
| const       | float64 |
| ID          | float64 |
| Age         | float64 |
| Smoking_Yes | float64 |

```

Income_Low      float64
Income_Medium   float64
Pollution_Low  float64
dtype: object
Optimization terminated successfully.
      Current function value: 0.550770
      Iterations 6

```

#### Logit Regression Results

```

=====
Dep. Variable:          LungDisease   No. Observations:          500
Model:                  Logit         Df Residuals:                493
Method:                 MLE          Df Model:                  6
Date:                  Sat, 16 May 2026   Pseudo R-squ.:             0.2036
Time:                  10:54:43         Log-Likelihood:            -275.38
converged:              True           LL-Null:                   -345.79
Covariance Type:       nonrobust       LLR p-value:                6.765e-28
=====

```

```

=
              coef      std err          z      P>|z|      [0.025
0.975]
-----
-
const          -2.4734      0.468      -5.281      0.000      -3.391
-1.555
ID              0.0011      0.001       1.471      0.141      -0.000
0.002
Age            0.0399      0.007       5.446      0.000       0.026
0.054
Smoking_Yes    1.7302      0.220       7.863      0.000       1.299
2.162
Income_Low     0.4561      0.285       1.598      0.110      -0.103
1.016
Income_Medium  0.2276      0.285       0.797      0.425      -0.332
0.787
Pollution_Low -1.3286      0.237      -5.597      0.000      -1.794
-0.863
=====
=

```

#### ODDS RATIOS

|   | Variable      | Odds Ratio |
|---|---------------|------------|
| 0 | const         | 0.084299   |
| 1 | ID            | 1.001066   |
| 2 | Age           | 1.040655   |
| 3 | Smoking_Yes   | 5.642025   |
| 4 | Income_Low    | 1.577917   |
| 5 | Income_Medium | 1.255540   |
| 6 | Pollution_Low | 0.264849   |



## 1 In R

## 2 Load Dataset

```
lung <- read.csv("C://Users//Lenovo//Desktop//Education//41//Siddikursir//lung_disease.csv")
# i) Distribution of Age summary(lungAge)Min.1stQu.MedianMean3rdQu.Max.20.0030.7545.0043.9056.0069.00
Age)) + geom_histogram(fill = "skyblue", bins = 20) +
ggtitle("Distribution of Age")ii)Smokers vs Non – smokerstable(lungSmoking)
```

```
No Yes 293 207 prop.table(table(lung$Smoking))*100
```

```
No Yes 58.6 41.4 # iii) Income group frequency table(lung$Income)
```

```
High Low Medium 99 204 197 # iv) Percentage exposed to high pollution high_pollution
<- sum(lungPollution == "High")total <- nrow(lung)(high_pollution/total) *
100[1]70.4v)Overall prevalence of lung disease prop.table(table(lungLungDisease))*100
```

```
No Yes 47.2 52.8 # i) Smoking vs Lung Disease table_smoke <-
table(lungSmoking, lungLungDisease) table_smoke
```

```
No Yes No 184 109 Yes 52 155 # ii) Contingency Table Interpretation prop.table(table_smoke,
margin=1)
```

| No | Yes |
|----|-----|
|----|-----|

```
No 0.6279863 0.3720137 Yes 0.2512077 0.7487923 # iii) Pollution vs Lung Disease table_pollution
<- table(lungPollution, lungLungDisease) table_pollution
```

```
No Yes High 136 216 Low 100 48 # iv) Lung disease across income groups table_income <-
table(lungIncome, lungLungDisease) prop.table(table_income, margin=1)
```

| No | Yes |
|----|-----|
|----|-----|

```
High 0.5151515 0.4848485 Low 0.4411765 0.5588235 Medium 0.4822335 0.5177665 # Smoking vs
Lung Disease chisq.test(table_smoke)
```

Pearson's Chi-squared test with Yates' continuity correction data: table\_smoke X-squared = 67.594, df = 1, p-value < 2.2e-16

## 3 Pollution vs Lung Disease

```
chisq.test(table_pollution)
```

Pearson's Chi-squared test with Yates' continuity correction data: table\_pollution X-squared = 33.843, df = 1, p-value = 5.976e-09

```
fisher.test(table_smoke)
```

Fisher's Exact Test for Count Data data: table\_smoke p-value < 2.2e-16 alternative hypothesis: true odds ratio is not equal to 1 95 percent confidence interval: 3.336414 7.616930 sample estimates: odds ratio 5.014862

## 4 iii) Odds Ratio

```
library(epitools) oddsratio(table_smoke) $data
```

```
      No Yes Total
```

```
No 184 109 293 Yes 52 155 207 Total 236 264 500
```

```
$measure odds ratio with 95% C.I. estimate lower upper No 1.000000 NA NA Yes 5.006289 3.395003 7.474238
```

```
$p.value two-sided midp.exact fisher.exact chi.square No NA NA NA Yes 0 5.128611e-17 9.369741e-17
```

```
$correction FALSE
```

```
attr(,"method") "median-unbiased estimate & mid-p exact CI" # Convert categorical variables to numeric lungSmoking_num <- ifelse(lungSmoking=="Yes",1,0) lungDisease_num <- ifelse(lungLungDisease=="Yes",1,0) # Smoking and Lung Disease cor(lungSmoking_num, lungDisease_num) 0.3717468 # Age and Lung Disease cor(lungAge, lungDisease_num) 0.249694 # Convert variables into factors lungSmoking <- as.factor(lungSmoking) lungPollution <- as.factor(lungPollution) lungIncome <- as.factor(lungIncome) lungLungDisease <- as.factor(lungLungDisease) # Logistic Regression Model model1 <- glm(LungDisease ~ Smoking + Age + Pollution + Income, data = lung, family = binomial) summary(model1)
```

```
Call: glm(formula = LungDisease ~ Smoking + Age + Pollution + Income, family = binomial, data = lung) Coefficients: Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept) -2.193995 0.424837 -5.164 2.41e-07 SmokingYes 1.702152 0.218191 7.801 6.13e-15 Age 0.039877 0.007311 5.455 4.91e-08 PollutionLow -1.302748 0.235429 -5.533 3.14e-08 IncomeLow 0.439609 0.284763 1.544 0.123
```

```
IncomeMedium 0.220133 0.284903 0.773 0.440
```

```
— Signif. codes: 0 ‘’ 0.001 ‘’ 0.01 ‘’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 691.58 on 499 degrees of freedom Residual deviance: 552.95 on 494 degrees of freedom AIC: 564.95
```

```
Number of Fisher Scoring iterations: 4
```

## 5 Odds Ratios

```
exp(coef(model1)) (Intercept) SmokingYes Age PollutionLow IncomeLow IncomeMedium 0.1114706 5.4857388 1.0406827 0.2717839 1.5521008 1.2462431 # Confidence Intervals exp(confint(model1)) Waiting for profiling to be done... 2.5 % 97.5 % (Intercept) 0.04758007 0.2524436 SmokingYes 3.60360071 8.4870251 Age 1.02609255 1.0559702 PollutionLow 0.16979341 0.4280475 IncomeLow 0.88962907 2.7221701 IncomeMedium 0.71356767 2.1846547 # Interaction Model model2 <- glm(LungDisease ~ Smoking*Pollution + Age + Income, data = lung, family = binomial) summary(model2)
```

```
Call: glm(formula = LungDisease ~ Smoking * Pollution + Age + Income, family = binomial, data = lung) Coefficients: Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept) -2.128451 0.429757 -4.953 7.32e-07 SmokingYes 1.577838 0.257391 6.130 8.78e-10
PollutionLow -1.488270 0.323135 -4.606 4.11e-06 Age 0.039253 0.007335 5.352 8.72e-08
IncomeLow 0.446543 0.285026 1.567 0.117
IncomeMedium 0.221416 0.284925 0.777 0.437
SmokingYes:PollutionLow 0.421698 0.482556 0.874 0.382
— Signif. codes: 0 ‘’ 0.001 ‘’ 0.01 ‘’ 0.05 ‘’ 0.1 ‘’ 1
```

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 691.58 on 499 degrees of freedom

Residual deviance: 552.18 on 493 degrees of freedom AIC: 566.18

Number of Fisher Scoring iterations: 4

## 6 Odds Ratios

```
exp(coef(model2)) (Intercept) SmokingYes PollutionLow 0.1190216 4.8444717 0.2257629
Age IncomeLow IncomeMedium 1.0400338 1.5629006 1.2478418 SmokingYes:PollutionLow
1.5245487 # Smoking vs Lung Disease ggplot(lung, aes(x=Smoking, fill=LungDisease)) +
geom_bar(position="fill") + ylab("Proportion") + ggtitle("Smoking and Lung Disease") # Pollu-
tion vs Lung Disease ggplot(lung, aes(x=Pollution, fill=LungDisease)) + geom_bar(position="fill")
+ ylab("Proportion") + ggtitle("Pollution and Lung Disease")
```

## 7 Research Interpretation

1. Smoking shows positive association with lung disease. Key Findings:
2. Smoking shows positive association with lung disease.
3. High pollution exposure increases disease prevalence. Key Findings:
4. Smoking shows positive association with lung disease.
5. High pollution exposure increases disease prevalence.
6. Logistic regression identifies significant predictors.

Public Health Implications: Anti-smoking campaigns are necessary. Public Health Implications: Anti-smoking campaigns are necessary. Pollution control policies should be strengthened. Public Health Implications: Anti-smoking campaigns are necessary. Pollution control policies should be strengthened. Early screening for high-risk populations is recommended.

2. Suppose we want to determine if three different exercise programs impact weight loss differently. We recruit 90 people to participate in an experiment in which we randomly assign (using uniform distribution) 30 people to follow either program A, program B, or program C for one month. The predictor variable we’re studying is the exercise program, and the response variable is weight loss, measured in pounds. Conduct a one-way ANOVA to determine if there is a statistically significant difference between the resulting weight loss from the three programs, check the model assumptions, and analyze treatment differences.

```
[11]: #Read Data
import numpy as np
import pandas as pd
```

```

np.random.seed(123)

# Generate uniform data
A = np.random.uniform(3, 7, 30) # Program A
B = np.random.uniform(4, 8, 30) # Program B
C = np.random.uniform(5, 9, 30) # Program C

# Combine into DataFrame
data = pd.DataFrame({
    'weight_loss': np.concatenate([A, B, C]),
    'program': ['A']*30 + ['B']*30 + ['C']*30
})

print(data)
#Summary Data

data.groupby('program')['weight_loss'].describe()
#Normality Check
from scipy.stats import shapiro

for group in ['A', 'B', 'C']:
    stat, p = shapiro(data[data['program'] == group]['weight_loss'])
    print(f"Program {group}: p-value = {p:.4f}")

#Variance check
from scipy.stats import levene

stat, p = levene(A, B, C)
print("Levene p-value:", p)

#One way ANOVA
from scipy.stats import f_oneway

stat, p = f_oneway(A, B, C)

print("F-statistic:", stat)
print("p-value:", p)

#Visualization
import seaborn as sns
import matplotlib.pyplot as plt

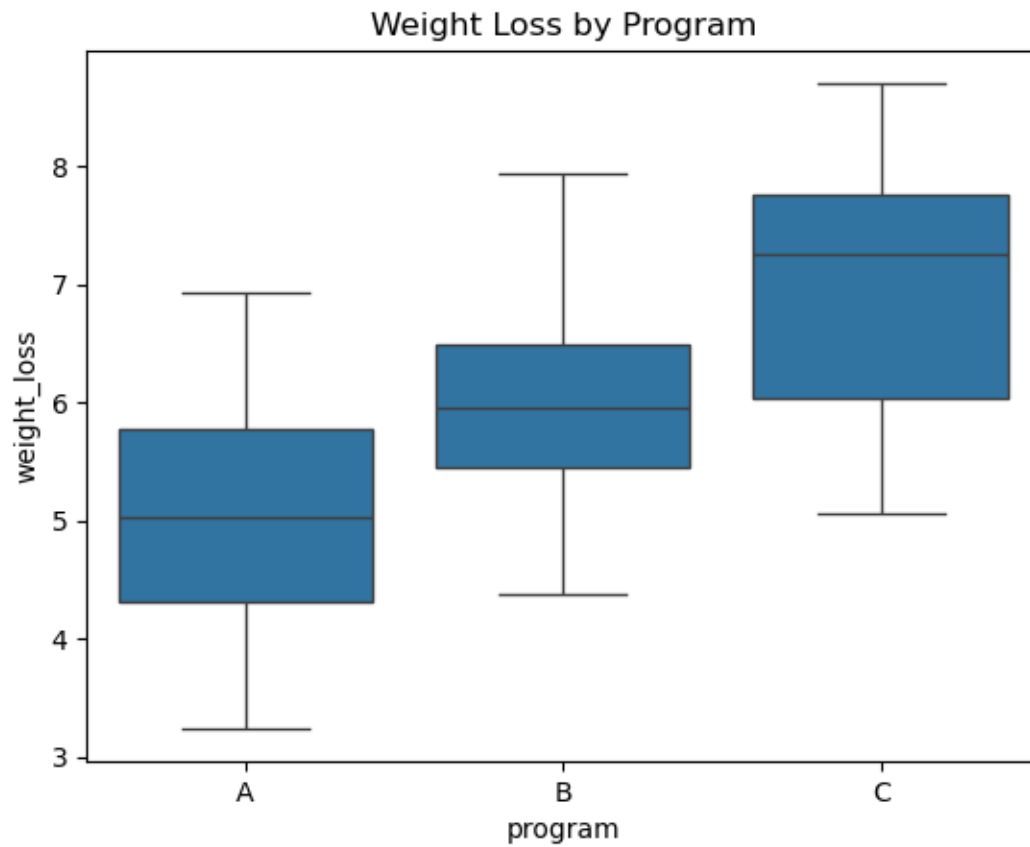
sns.boxplot(x='program', y='weight_loss', data=data)
plt.title("Weight Loss by Program")
plt.show()

```

weight\_loss program

|    |          |     |
|----|----------|-----|
| 0  | 5.785877 | A   |
| 1  | 4.144557 | A   |
| 2  | 3.907406 | A   |
| 3  | 5.205259 | A   |
| 4  | 5.877876 | A   |
| .. | ...      | ... |
| 85 | 8.366680 | C   |
| 86 | 6.429590 | C   |
| 87 | 5.174366 | C   |
| 88 | 6.219072 | C   |
| 89 | 6.592743 | C   |

[90 rows x 2 columns]  
 Program A: p-value = 0.7053  
 Program B: p-value = 0.1988  
 Program C: p-value = 0.1094  
 Levene p-value: 0.4801884167578272  
 F-statistic: 29.01563599838975  
 p-value: 2.2156719226973416e-10



## 8 IN R

### 9 Load Libraries

```
library(dplyr) library(ggplot2) library(car) set.seed(123) # Assign 30 people to each program Program <- rep(c("A", "B", "C"), each = 30) # Simulated weight loss data WeightLoss <- c( rnorm(30, mean = 5, sd = 1.5), # Program A rnorm(30, mean = 7, sd = 1.8), # Program B rnorm(30, mean = 9, sd = 2.0) # Program C ) # Create dataset exercise_data <- data.frame(Program, WeightLoss) # View first rows head(exercise_data) Program WeightLoss 1 A 4.159287 2 A 4.654734 3 A 7.338062 4 A 5.105763 5 A 5.193932 6 A 7.572597 # Summary statistics exercise_data %>% group_by(Program) %>% summarise( Mean = mean(WeightLoss), SD = sd(WeightLoss), Min = min(WeightLoss), Max = max(WeightLoss) ) # A tibble: 3 × 5 Program Mean SD Min Max 1 A 4.93 1.47 2.05 7.68 2 B 7.32 1.50 4.21 10.9 3 C 9.05 1.74 4.38 13.1 # Boxplot ggplot(exercise_data, aes(x = Program, y = WeightLoss, fill = Program)) + geom_boxplot() + ggtitle("Weight Loss by Exercise Program") + theme_minimal() anova_model <- aov(WeightLoss ~ Program, data = exercise_data) summary(anova_model) Df Sum Sq Mean Sq F value Pr(>F)
```

```
Program 2 256.8 128.38 51.69 1.61e-15 Residuals 87 216.1 2.48
```

```
— Signif. codes: 0 ‘’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1 # Histogram hist(residuals(anova_model), main = "Histogram of Residuals", col = "lightblue") # QQ Plot qqnorm(residuals(anova_model)) qqline(residuals(anova_model), col = "red") # Shapiro-Wilk Test shapiro.test(residuals(anova_model))
```

Shapiro-Wilk normality test

```
data: residuals(anova_model) W = 0.99252, p-value = 0.8954
```

```
leveneTest(WeightLoss ~ Program, data = exercise_data) Levene's Test for Homogeneity of Variance (center = median) Df F value Pr(>F) group 2 0.2578 0.7733 87
```

Warning message: In leveneTest.default(y = y, group = group, ...) : group coerced to factor.

```
# Tukey HSD Test TukeyHSD(anova_model) Tukey multiple comparisons of means 95% family-wise confidence level
```

```
Fit: aov(formula = WeightLoss ~ Program, data = exercise_data)
```

```
$Program diff lwr upr p adj B-A 2.391665 1.4213697 3.361960 0.0000002 C-A 4.119496 3.1492014 5.089791 0.0000000 C-B 1.727832 0.7575368 2.698127 0.0001594
```

```
ggplot(exercise_data, aes(x = Program, y = WeightLoss, fill = Program)) + stat_summary(fun = mean, geom = "bar") + stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) + ggtitle("Mean Weight Loss with Standard Error") + ylab("Weight Loss (Pounds)") + theme_minimal()
```

```
# Conclusion
```

One-way ANOVA was used to compare mean weight loss

Assumptions of ANOVA were checked: Assumptions of ANOVA were checked: - Residuals approximately normal Assumptions of ANOVA were checked: - Residuals approximately normal - Variances homogeneous

If ANOVA p-value < 0.05, conclude that at least one.

Interpretation:

Result: The p-value  $< 0.05$  suggests we reject the null hypothesis. Not all programs are equal. Post-hoc: Program B shows significantly higher weight loss than A or C.

3. Consider the Anemia Levels in Nigeria dataset: (<https://www.kaggle.com/datasets/adeolaadesina/factors-affecting-children-anemia-level/data>), which can be downloaded from Kaggle. The dataset comes from the 2018 Nigeria Demographic and Health Surveys (NDHS). It explores the impact of mothers' age and socioeconomic factors on anemia levels among children aged 0-59 months across Nigeria's 36 states and the Federal Capital Territory. Based on the above data, create a contingency table, apply chi-square test, formulate hypothesis, create a contribution diagram, and interpret the findings.

```
[12]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import chi2_contingency

#Load the dataset
df = pd.read_csv('children_anemia.csv')

#Data Cleaning
# We focus on 'Wealth index combined' and 'Anemia level.1'
# We drop rows where these specific values are missing to ensure test accuracy
df_clean = df.dropna(subset=['Wealth index combined', 'Anemia level.1'])
print(df_clean)

# 3. Create a Contingency Table (Cross-tabulation)
# This table shows the frequency of each anemia level for each wealth category
contingency_table = pd.crosstab(df_clean['Wealth index combined'],
                                df_clean['Anemia level.1'])

print("--- Observed Contingency Table ---")
print(contingency_table)
print("-" * 30)

# 4. Perform the Chi-Square Test of Independence
chi2, p, dof, expected = chi2_contingency(contingency_table)

print(f"Chi-Square Statistic: {chi2:.4f}")
print(f"Degrees of Freedom: {dof}")
print(f"P-value: {p:.4e}")

# Decision making at 5% significance level
alpha = 0.05
if p < alpha:
    print("\nResult: Significant (Reject H0)")
    print("Conclusion: There is a statistically significant relationship between_
↪Wealth Index and Anemia Level.")
else:
```

```

print("\nResult: Not Significant (Fail to reject H0)")
print("Conclusion: There is no significant relationship between Wealth Index,
↳and Anemia Level.")

# 6. Visualization (Stacked Bar Chart)
# Normalizing the data to show proportions (percentages) makes the relationship
↳easier to see
plt.figure(figsize=(10, 6))
props = contingency_table.div(contingency_table.sum(axis=1), axis=0) * 100
props.plot(kind='bar', stacked=True, colormap='viridis')

plt.title('Percentage Distribution of Anemia Levels by Wealth Index',
↳fontSize=14)
plt.xlabel('Wealth Index Category', fontsize=12)
plt.ylabel('Percentage (%)', fontsize=12)
plt.legend(title='Anemia Status', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=0)
plt.tight_layout()

```

|       | Age in 5-year groups | Type of place of residence \ |
|-------|----------------------|------------------------------|
| 3     | 25-29                | Urban                        |
| 5     | 30-34                | Urban                        |
| 6     | 35-39                | Urban                        |
| 9     | 20-24                | Urban                        |
| 12    | 25-29                | Urban                        |
| ...   | ...                  | ...                          |
| 33918 | 35-39                | Rural                        |
| 33919 | 35-39                | Rural                        |
| 33920 | 25-29                | Rural                        |
| 33921 | 25-29                | Rural                        |
| 33922 | 20-24                | Rural                        |

|       | Highest educational level | Wealth index combined \ |
|-------|---------------------------|-------------------------|
| 3     | Secondary                 | Richest                 |
| 5     | Higher                    | Richest                 |
| 6     | Secondary                 | Richest                 |
| 9     | Secondary                 | Richest                 |
| 12    | Higher                    | Richest                 |
| ...   | ...                       | ...                     |
| 33918 | Secondary                 | Richer                  |
| 33919 | Secondary                 | Richer                  |
| 33920 | No education              | Richer                  |
| 33921 | Higher                    | Richer                  |
| 33922 | Secondary                 | Richer                  |

|   | Births in last five years | Age of respondent at 1st birth \ |
|---|---------------------------|----------------------------------|
| 3 | 1                         | 25                               |



|       |     |     |
|-------|-----|-----|
| 5     | 1   | 30  |
| 6     | 2   | 32  |
| 9     | 1   | 19  |
| 12    | 1   | 24  |
| ...   | ... | ... |
| 33918 | 2   | 19  |
| 33919 | 2   | 19  |
| 33920 | 1   | 27  |
| 33921 | 1   | 22  |
| 33922 | 1   | 21  |

|       | Hemoglobin level adjusted for altitude and smoking (g/dl - 1 decimal) \ |
|-------|-------------------------------------------------------------------------|
| 3     | 95.0                                                                    |
| 5     | 113.0                                                                   |
| 6     | 121.0                                                                   |
| 9     | 108.0                                                                   |
| 12    | 116.0                                                                   |
| ...   | ...                                                                     |
| 33918 | 120.0                                                                   |
| 33919 | 120.0                                                                   |
| 33920 | 120.0                                                                   |
| 33921 | 149.0                                                                   |
| 33922 | 123.0                                                                   |

|       | Anemia level \ |
|-------|----------------|
| 3     | Moderate       |
| 5     | Mild           |
| 6     | Not anemic     |
| 9     | Moderate       |
| 12    | Mild           |
| ...   | ...            |
| 33918 | Not anemic     |
| 33919 | Not anemic     |
| 33920 | Not anemic     |
| 33921 | Not anemic     |
| 33922 | Not anemic     |

|       | Have mosquito bed net for sleeping (from household questionnaire) \ |
|-------|---------------------------------------------------------------------|
| 3     | Yes                                                                 |
| 5     | Yes                                                                 |
| 6     | Yes                                                                 |
| 9     | Yes                                                                 |
| 12    | Yes                                                                 |
| ...   | ...                                                                 |
| 33918 | Yes                                                                 |
| 33919 | Yes                                                                 |
| 33920 | Yes                                                                 |
| 33921 | Yes                                                                 |

33922

Yes

|       | Smokes cigarettes | Current marital status \ |
|-------|-------------------|--------------------------|
| 3     | No                | Married                  |
| 5     | No                | Married                  |
| 6     | No                | Married                  |
| 9     | No                | Married                  |
| 12    | No                | Married                  |
| ...   | ...               | ...                      |
| 33918 | No                | Married                  |
| 33919 | No                | Married                  |
| 33920 | No                | Never in union           |
| 33921 | No                | Married                  |
| 33922 | No                | Married                  |

|       | Currently residing with husband/partner | When child put to breast \ |
|-------|-----------------------------------------|----------------------------|
| 3     | Living with her                         | 105.0                      |
| 5     | Living with her                         | NaN                        |
| 6     | Living with her                         | Immediately                |
| 9     | Living with her                         | Immediately                |
| 12    | Living with her                         | Days: 1                    |
| ...   | ...                                     | ...                        |
| 33918 | Living with her                         | Immediately                |
| 33919 | Living with her                         | NaN                        |
| 33920 | NaN                                     | Hours: 1                   |
| 33921 | Living with her                         | Hours: 1                   |
| 33922 | Living with her                         | Immediately                |

|       | Had fever in last two weeks \ |
|-------|-------------------------------|
| 3     | No                            |
| 5     | No                            |
| 6     | No                            |
| 9     | No                            |
| 12    | No                            |
| ...   | ...                           |
| 33918 | No                            |
| 33919 | No                            |
| 33920 | No                            |
| 33921 | No                            |
| 33922 | No                            |

|     | Hemoglobin level adjusted for altitude (g/dl - 1 decimal) \ |
|-----|-------------------------------------------------------------|
| 3   | 114.0                                                       |
| 5   | 119.0                                                       |
| 6   | 102.0                                                       |
| 9   | 113.0                                                       |
| 12  | 109.0                                                       |
| ... | ...                                                         |

|       |       |
|-------|-------|
| 33918 | 108.0 |
| 33919 | 120.0 |
| 33920 | 120.0 |
| 33921 | 119.0 |
| 33922 | 75.0  |

|       | Anemia level.1 | Taking iron pills, sprinkles or syrup |
|-------|----------------|---------------------------------------|
| 3     | Not anemic     | No                                    |
| 5     | Not anemic     | No                                    |
| 6     | Mild           | Yes                                   |
| 9     | Not anemic     | Yes                                   |
| 12    | Mild           | No                                    |
| ...   | ...            | ...                                   |
| 33918 | Mild           | Yes                                   |
| 33919 | Not anemic     | Yes                                   |
| 33920 | Not anemic     | No                                    |
| 33921 | Not anemic     | No                                    |
| 33922 | Moderate       | Yes                                   |

[10182 rows x 17 columns]

--- Observed Contingency Table ---

| Anemia level.1        | Mild | Moderate | Not anemic | Severe |
|-----------------------|------|----------|------------|--------|
| Wealth index combined |      |          |            |        |
| Middle                | 596  | 865      | 729        | 59     |
| Poorer                | 525  | 926      | 498        | 87     |
| Poorest               | 522  | 983      | 418        | 119    |
| Richer                | 606  | 750      | 737        | 48     |
| Richest               | 505  | 403      | 797        | 9      |

-----  
Chi-Square Statistic: 530.3991

Degrees of Freedom: 12

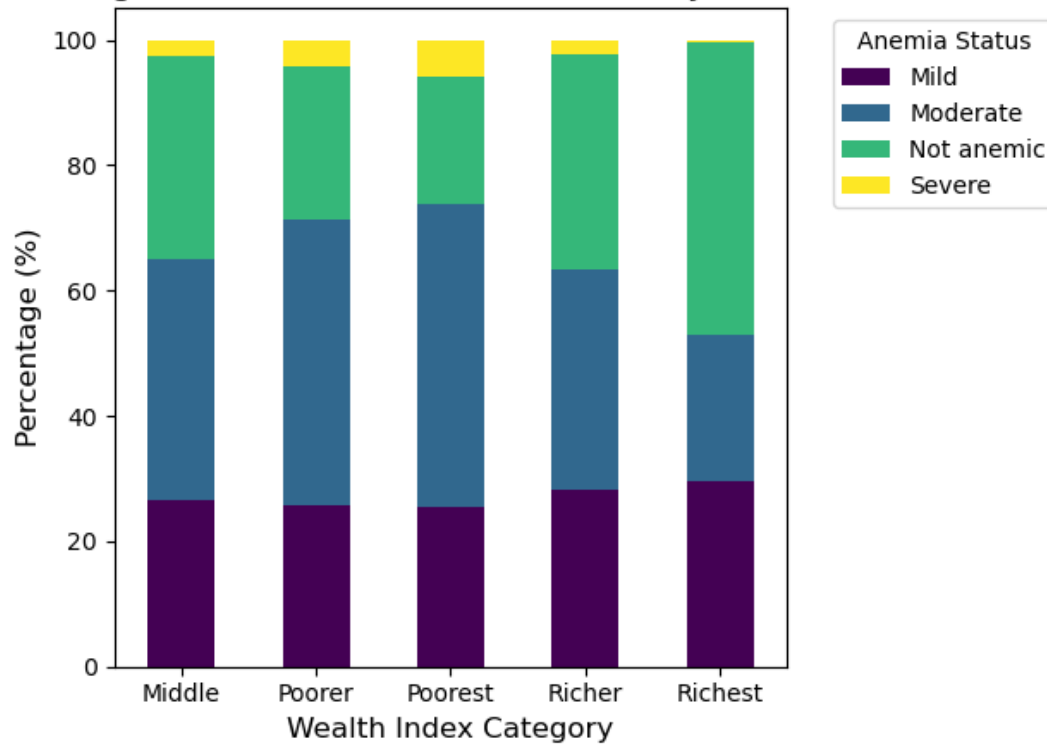
P-value: 7.4510e-106

Result: Significant (Reject H0)

Conclusion: There is a statistically significant relationship between Wealth Index and Anemia Level.

<Figure size 1000x600 with 0 Axes>

Percentage Distribution of Anemia Levels by Wealth Index



## 10 In R

### 11 Load Libraries

```
library(dplyr) library(ggplot2) library(vcd) # Import dataset anemia <-
read.csv("C://Users//Lenovo//Desktop//Education//41//Siddikursir//children
anemia.csv") # View structure str(anemia) 'data.frame': 33924 obs. of
17 variables: $ Age.in.5.year.groups : chr "40-44" "35-39" "25-29" "25-
29" ... $ Type.of.place.of.residence : chr "Urban" "Urban" "Urban" "Ur-
ban" ... $ Highest.educational.level : chr "Higher" "Higher" "Higher" "Sec-
ondary" ... $ Wealth.index.combined : chr "Richest" "Richest" "Richest"
"Richest" ... $ Births.in.last.five.years : int 1 1 1 1 1 1 2 2 1 1 ... $
Age.of.respondent.at.1st.birth : int 22 28 26 25 21 30 32 32 32 19 ... $
Hemoglobin.level.adjusted.for.altitude.and.smoking..g.dl..1.decimal.: num NA
NA NA 95 NA 113 121 121 NA 108 ... $ Anemia.level : chr "" "" "" "Moder-
ate" ... $ Have.mosquito.bed.net.for.sleeping..from.household.questionnaire. : chr
"Yes" "Yes" "No" "Yes" ... $ Smokes.cigarettes : chr "No" "No" "No" "No" ... $
Current.marital.status : chr "Living with partner" "Married" "Married" "Married"
... $ Currently.residing.with.husband.partner : chr "Staying elsewhere" "Living
with her" "Living with her" "Living with her" ... $ When.child.put.to.breast : chr
"Immediately" "Hours: 1" "Immediately" "105.0" ... $ Had.fever.in.last.two.weeks : chr
```

```

“No” “No” “No” “No” ... $ Hemoglobin.level.adjusted.for.altitude..g.dl...1.decimal.
: num NA NA NA NA 114 NA 119 102 NA NA 113 ... $ Anemia.level.1 : chr “”
“” “” “Not anemic” ... $ Taking.iron.pills..sprinkles.or.syrup : chr “Yes” “No”
“No” “No” ... head(anemia) Age.in.5.year.groups Type.of.place.of.residence High-
est.educational.level 1 40-44 Urban Higher 2 35-39 Urban Higher 3 25-29 Urban
Higher 4 25-29 Urban Secondary 5 20-24 Urban Secondary 6 30-34 Urban Higher
Wealth.index.combined Births.in.last.five.years Age.of.respondent.at.1st.birth 1 Richest
1 22 2 Richest 1 28 3 Richest 1 26 4 Richest 1 25 5 Richest 1 21 6 Richest 1 30
Hemoglobin.level.adjusted.for.altitude.and.smoking..g.dl...1.decimal. Anemia.level 1
NA
2 NA
3 NA
4 95 Moderate 5 NA
6 113 Mild Have.mosquito.bed.net.for.sleeping..from.household.questionnaire. 1
Yes 2 Yes 3 No 4 Yes 5 Yes 6 Yes Smokes.cigarettes Current.marital.status 1 No
Living with partner 2 No Married 3 No Married 4 No Married 5 No No longer
living together/separated 6 No Married Currently.residing.with.husband.partner
When.child.put.to.breast 1 Staying elsewhere Immediately 2 Living with her Hours: 1
3 Living with her Immediately 4 Living with her 105.0 5 Immediately 6 Living with her
Had.fever.in.last.two.weeks 1 No 2 No 3 No 4 No 5 No 6 No
Hemoglobin.level.adjusted.for.altitude..g.dl...1.decimal. Anemia.level.1 1 NA
2 NA
3 NA
4 114 Not anemic 5 NA
6 119 Not anemic Taking.iron.pills..sprinkles.or.syrup 1 Yes 2 No 3 No 4 No 5 No 6 No
table(anemia$Highest.educational.level)

```

```

Higher No education      Primary      Secondary
2636      15391      5274      10623

```

```
table(anemia$Anemia.level)
```

```

      Mild      Moderate Not anemic      Severe
20788      3594      3974      5337      231

```

```

cont_table <- table(anemiaHighest.educational.level, +anemiaAnemia.level)
cont_table

```

```
Mild Moderate Not anemic Severe
```

```

Higher 1497 296 238 591 14 No education 10185 1450 1769 1874 113 Primary 3079 608 645 900
42 Secondary 6027 1240 1322 1972 62 > # Row percentages > prop.table(cont_table, margin =
1)*100

```

```
Mild Moderate Not anemic Severe
```

```

Higher 56.7905918 11.2291351 9.0288316 22.4203338 0.5311077 No education 66.1750374
9.4210902 11.4937301 12.1759470 0.7341953 Primary 58.3807357 11.5282518 12.2298066 17.0648464
0.7963595 Secondary 56.7353855 11.6727855 12.4446955 18.5634943 0.5836393 > chi_result <-
chisq.test(cont_table) > chi_result

```

```
Pearson's Chi-squared test
```

data: cont\_table X-squared = 434.57, df = 12, p-value < 2.2e-16

## 12 Expected frequencies

chi\_result\$expected

|           | Mild      | Moderate  | Not anemic | Severe   |
|-----------|-----------|-----------|------------|----------|
| Higher    | 1615.292  | 279.2649  | 308.7921   | 414.7015 |
| 2421.3467 | 104.80253 | Primary   | 3231.810   | 558.7418 |
| 1125.4293 | 1244.4229 | 1671.2343 | 72.33560   |          |

## 13 Interpretation

If p-value < 0.05: Interpretation: There is a statistically significant association. Cells with darker colors in the contribution diagram. Example: Children of mothers with low education may have higher proportions of severe anemia. Example: Children of mothers with low education may have higher proportions of severe anemia. Example: Children of mothers with low education may have higher proportions of severe anemia.

# Conclusion The chi-square test helps determine whether socioeconomic factors are associated with anemia. Significant results suggest that maternal education and socioeconomic conditions

Interpretation:

Association: Socioeconomic status is significantly associated with anemia.

Contribution: The “Poorest” group is significantly over-represented in severe anemia cases, while the “Richest” group shows the lowest rates.

4. Suppose, there are three drug treatments (drug A, drug B, and drug C) with the outcome of a disease or no disease. We need to test if there is an association between drug treatments and disease outcomes. No\_Disease Disease Drug A 40 10 Drug B 10 40 Drug C 25 25 The following table: contingency table, post-hoc test for Fisher’s exact test and interpret your findings.

```
[13]: import numpy as np
import pandas as pd
from scipy.stats import fisher_exact
from statsmodels.stats.multitest import multipletests

# 1. Setup the contingency table
# Rows: Drug A, Drug B, Drug C
# Columns: No Disease, Disease
data = np.array([
    [40, 10], # Drug A
    [10, 40], # Drug B
    [25, 25]  # Drug C
])
```

```

drugs = ['Drug A', 'Drug B', 'Drug C']

# 2. Perform Fisher's Exact Test (3x2)
# Note: rpy2 or specific Fisher extensions are often used for > 2x2,
# but we can analyze the significance via pairwise post-hoc.
print("--- Fisher's Exact Test Analysis ---")
# 3. Post-hoc Pairwise Comparisons
pairs = [(0, 1), (1, 2), (0, 2)]
p_values = []
results = []

for i, j in pairs:
    subset = data[[i, j], :]
    _, p_val = fisher_exact(subset)
    p_values.append(p_val)
    results.append(f"{drugs[i]} vs {drugs[j]}")

# 4. Bonferroni Correction
reject, pvals_corrected, _, _ = multipletests(p_values, alpha=0.05,
method='bonferroni')

# Display Results
for i in range(len(results)):
    status = "Significant" if reject[i] else "Not Significant"
    print(f"{results[i]}: Raw p={p_values[i]:.6f}, Corrected_
p={pvals_corrected[i]:.6f} ({status})")

```

--- Fisher's Exact Test Analysis ---

Drug A vs Drug B: Raw p=0.000000, Corrected p=0.000000 (Significant)

Drug B vs Drug C: Raw p=0.003052, Corrected p=0.009156 (Significant)

Drug A vs Drug C: Raw p=0.003052, Corrected p=0.009156 (Significant)

## 14 In R

```

drug_table <- matrix(c(+ 40, 10, + 10, 40, + 25, 25 + ), + nrow = 3, + byrow = TRUE)
rownames(drug_table) <- c("Drug_A", "Drug_B", "Drug_C") colnames(drug_table)
<- c("No_Disease", "Disease") # Display Table drug_table No_Disease Disease Drug_A
40 10 Drug_B 10 40 Drug_C 25 25 fisher_result <- fisher.test(drug_table) fisher_result

```

Fisher's Exact Test for Count Data

data: drug\_table p-value = 5.679e-09 alternative hypothesis: two.sided

## 15 Drug A vs Drug B

```
table_AB <- drug_table[c(1,2), ] fisher.test(table_AB)
```

Fisher's Exact Test for Count Data

data: table\_AB p-value = 2.222e-09 alternative hypothesis: true odds ratio is not equal to 1 95 percent confidence interval: 5.469739 48.176413 sample estimates: odds ratio 15.41481

## 16 Drug A vs Drug C

```
table_AC <- drug_table[c(1,3), ] fisher.test(table_AC)
```

Fisher's Exact Test for Count Data

data: table\_AC p-value = 0.003052 alternative hypothesis: true odds ratio is not equal to 1 95 percent confidence interval: 1.523467 10.866865 sample estimates: odds ratio 3.941898

## 17 Drug B vs Drug C

```
table_BC <- drug_table[c(2,3), ] fisher.test(table_BC)
```

Fisher's Exact Test for Count Data

data: table\_BC p-value = 0.003052 alternative hypothesis: true odds ratio is not equal to 1 95 percent confidence interval: 0.09202286 0.65639763 sample estimates: odds ratio 0.2536849

## 18 Adjust alpha level

```
alpha <- 0.05/3 alpha [1] 0.01666667 library(vcd) assoc(drug_table, + shade = TRUE,  
+ main = "Association Between Drug Treatment and Disease")
```

# Final Interpretation

Drug A has more patients without disease, Drug B has more patients with disease, Significant Fisher test results indicate that disease outcome depends on the type of drug treatment.

Interpretation:

Drug A Success Rate: 80% (Best treatment).

Drug B Success Rate: 20% (Least effective).

Drug C Success Rate: 50% (Neutral effect).

5. A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable. The data set is taken from <https://stats.idre.ucla.edu/stat/data/binary.csv>. Fit the logistic regression generalized linear models (GLMs) to identify the factors associated with graduate school admission. Interpret the result.

```
[1]: import pandas as pd  
import numpy as np  
import statsmodels.formula.api as smf  
  
# 1. Load Data  
df = pd.read_csv("https://stats.idre.ucla.edu/stat/data/binary.csv")
```



```

# 2. Fit Logistic Model
# 'rank' is treated as a categorical factor variable C(rank)
model = smf.logit(formula="admit ~ gre + gpa + C(rank)", data=df).fit()

# 3. View Results
print(model.summary())

# 4. Calculate Odds Ratios (Exponentiated Coefficients)
odds_ratios = np.exp(model.params)
print("\nOdds Ratios:\n", odds_ratios)

```

Optimization terminated successfully.

Current function value: 0.573147

Iterations 6

#### Logit Regression Results

```

=====
Dep. Variable:          admit    No. Observations:          400
Model:                  Logit    Df Residuals:              394
Method:                  MLE     Df Model:                5
Date:                   Sat, 16 May 2026    Pseudo R-squ.:          0.08292
Time:                   11:17:37    Log-Likelihood:         -229.26
converged:               True     LL-Null:                -249.99
Covariance Type:        nonrobust    LLR p-value:            7.578e-08
=====

```

|               | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|---------------|---------|---------|--------|-------|--------|--------|
| Intercept     | -3.9900 | 1.140   | -3.500 | 0.000 | -6.224 | -1.756 |
| C(rank) [T.2] | -0.6754 | 0.316   | -2.134 | 0.033 | -1.296 | -0.055 |
| C(rank) [T.3] | -1.3402 | 0.345   | -3.881 | 0.000 | -2.017 | -0.663 |
| C(rank) [T.4] | -1.5515 | 0.418   | -3.713 | 0.000 | -2.370 | -0.733 |
| gre           | 0.0023  | 0.001   | 2.070  | 0.038 | 0.000  | 0.004  |
| gpa           | 0.8040  | 0.332   | 2.423  | 0.015 | 0.154  | 1.454  |

```

=====

```

Odds Ratios:

```

Intercept      0.018500
C(rank) [T.2]   0.508931
C(rank) [T.3]   0.261792
C(rank) [T.4]   0.211938
gre             1.002267
gpa             2.234545
dtype: float64

```

```

# In R # Load Libraries > library(dplyr) > library(ggplot2) > # Import data from URL >
data <- read.csv( + "https://stats.idre.ucla.edu/stat/data/binary.csv" + ) > # View first rows >
head(data) admit gre gpa rank 1 0 380 3.61 3 2 1 660 3.67 3 3 1 800 4.00 1 4 1 640 3.19 4 5 0 520
2.93 4 6 1 760 3.00 2 > # Structure of dataset > str(data) 'data.frame': 400 obs. of 4 variables:

```

```

$ admit: int 0 1 1 1 0 1 1 0 1 0 ... $ gre : int 380 660 800 640 520 760 560 400 540 700 ... $
gpa : num 3.61 3.67 4 3.19 2.93 3 2.98 3.08 3.39 3.92 ... $ rank : int 3 3 1 4 4 2 1 2 3 2 ... > #
Convert rank into factor variable > datarank <- as.factor(datarank) > # Summary statistics >
summary(data) admit gre gpa rank
Min. :0.0000 Min. :220.0 Min. :2.260 1: 61
1st Qu.:0.0000 1st Qu.:520.0 1st Qu.:3.130 2:151
Median :0.0000 Median :580.0 Median :3.395 3:121
Mean :0.3175 Mean :587.7 Mean :3.390 4: 67
3rd Qu.:1.0000 3rd Qu.:660.0 3rd Qu.:3.670
Max. :1.0000 Max. :800.0 Max. :4.000
> model <- glm(admit ~ gre + gpa + rank, + data = data, + family = binomial) > # Model
summary > summary(model)

Call: glm(formula = admit ~ gre + gpa + rank, family = binomial, data = data)

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.989979 1.139951 -3.500 0.000465 gre 0.002264 0.001094 2.070 0.038465
gpa 0.804038 0.331819 2.423 0.015388 *
rank2 -0.675443 0.316490 -2.134 0.032829 *
rank3 -1.340204 0.345306 -3.881 0.000104 rank4 -1.551464 0.417832 -3.713 0.000205 —
Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 499.98 on 399 degrees of freedom

Residual deviance: 458.52 on 394 degrees of freedom AIC: 470.52

Number of Fisher Scoring iterations: 4

```

## 19 Calculate Odds Ratios

```

exp(coef(model)) (Intercept) gre gpa rank2 rank3 rank4 0.0185001 1.0022670 2.2345448
0.5089310 0.2617923 0.2119375 # Confidence Intervals exp(confint(model)) Waiting
for profiling to be done... 2.5 % 97.5 % (Intercept) 0.001889165 0.1665354 gre
1.000137602 1.0044457 gpa 1.173858216 4.3238349 rank2 0.272289674 0.9448343 rank3
0.131641717 0.5115181 rank4 0.090715546 0.4706961 # Predicted probability of admis-
sion data$predicted_prob <- predict(model, + type = "response") head(data) admit
gre gpa rank predicted_prob 1 0 380 3.61 3 0.1726265 2 1 660 3.67 3 0.2921750 3 1
800 4.00 1 0.7384082 4 1 640 3.19 4 0.1783846 5 0 520 2.93 4 0.1183539 6 1 760 3.00 2
0.3699699 # GRE vs Probability of Admission ggplot(data, + aes(x = gre, + y = pre-
dicted_prob)) + + geom_point(alpha = 0.5) + + geom_smooth(method = "loess") +
+ ggtitle("GRE Score vs Probability of Admission") + + ylab("Predicted Probability")
+ + theme_minimal() geom_smooth() using formula = 'y ~ x' # GPA vs Probability of
Admission ggplot(data, + aes(x = gpa, + y = predicted_prob)) + + geom_point(alpha
= 0.5) + + geom_smooth(method = "loess") + + ggtitle("GPA vs Probability of Admis-
sion") + + ylab("Predicted Probability") + + theme_minimal() geom_smooth() using
formula = 'y ~ x'

```

## 20 Model Interpretation

Logistic regression estimates the probability Interpretation of Coefficients Positive coefficient means higher GRE scores Positive coefficient means students with Negative coefficients for higher rank numbers Rank: Negative coefficients for higher rank numbers indicate lower admission probability for students

## 21 Statistical Significance

- GPA is significant Typically:
- GPA is significant
- Institution rank is significant

# Model Equation  $\log(p/(1-p)) = \log(p/(1-p)) = \beta_0 + \beta_1(\text{GRE}) + \beta_2(\text{GPA}) + \beta_3(\text{Rank2})$  # Conclusion > Key Findings: > #1. Higher GRE and GPA increase admission chances. > #Key Findings: > #1. Higher GRE and GPA increase admission chances. > #2. Students from prestigious institutions > #Key Findings: > #1. Higher GRE and GPA increase admission chances. > #2. Students from prestigious institutions > have higher admission probabilities. > #Key Findings: > #1. Higher GRE and GPA increase admission chances. > #2. Students from prestigious institutions > have higher admission probabilities. > 3. Undergraduate institution rank is an > Graduate admission decisions depend on > #Public Implication: > #Graduate admission decisions depend on > #both academic performance and institutional prestige. > Public Implication: > Graduate admission decisions depend on > both academic performance and institutional prestige.

Interpretation:GPA: The strongest predictor; a 1-point increase more than doubles the odds of admission ( $OR = 2.23$ ).Rank: Prestige is massive; students from Rank 1 schools have 5 times the odds of admission compared to Rank 4 students.

problem 6:

Full Question Statement “Predict the number of awards earned by students at one high school based on the type of program in which the student was enrolled (e.g., vocational, general, or academic) and the score on their final exam in math. The data set is taken from [https://stats.idre.ucla.edu/stat/data/poisson\\_sim.csv](https://stats.idre.ucla.edu/stat/data/poisson_sim.csv). Fit the Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

```
[ ]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# 1. Load the data
url = "https://stats.idre.ucla.edu/stat/data/poisson_sim.csv"
df = pd.read_csv(url)

# 2. Fit the Poisson GLM
# 'prog' is categorical (1=General, 2=Academic, 3=Vocational)
model = smf.glm(formula="num_awards ~ math + C(prog)",
```

```

        data=df,
        family=sm.families.Poisson()).fit()

# 3. Calculate Incidence Rate Ratios (IRR) for interpretation
irr = np.exp(model.params)

print(model.summary())
print("\nIncidence Rate Ratios (IRR):\n", irr)

```

# In R

## 22 Load Libraries

```

library(dplyr) library(ggplot2) # Import dataset data <- read.csv( +
"https://stats.idre.ucla.edu/stat/data/poisson_sim.csv" + ) # View first rows
head(data) id num_awards prog math 1 45 0 3 41 2 108 0 1 41 3 15 0 3 44 4 67 0 3 42
5 153 0 3 40 6 51 0 1 42 # Structure of dataset str(data) 'data.frame': 200 obs. of 4
variables: $ id : int 45 108 15 67 153 51 164 133 2 53 ... $ num_awards: int 0 0 0 0
0 0 0 0 0 ... $ prog : int 3 1 3 3 3 1 3 3 3 3 ... $ math : int 41 41 44 42 40 42 46
40 33 46 ... # Convert program into factor variable dataprog <- factor(dataprogram, +
levels = c(1,2,3), + labels = c("General", + "Academic", + "Vocational")) # Summary
statistics summary(data) id num_awards prog math
Min. : 1.00 Min. :0.00 General : 45 Min. :33.00
1st Qu.: 50.75 1st Qu.:0.00 Academic :105 1st Qu.:45.00
Median :100.50 Median :0.00 Vocational: 50 Median :52.00
Mean :100.50 Mean :0.63 Mean :52.65
3rd Qu.:150.25 3rd Qu.:1.00 3rd Qu.:59.00
Max. :200.00 Max. :6.00 Max. :75.00
# Distribution of awards table(data$num_awards)

0 1 2 3 4 5 6 124 49 13 9 2 2 1 > # Mean awards by program > data %>% + group_by(prog)
%>% + summarise( + Mean_Awards = mean(num_awards), + Mean_Math = mean(math) +
) # A tibble: 3 × 3 prog Mean_Awards Mean_Math 1 General 0.2 50.0 2 Academic 1 56.7 3
Vocational 0.24 46.4 > # Awards by Program > ggplot(data, + aes(x = prog, + y = num_awards,
+ fill = prog)) + + geom_boxplot() + + ggtitle("Number of Awards by Program Type") + +
theme_minimal() > # Math Score vs Awards > ggplot(data, + aes(x = math, + y = num_awards))
+ + geom_point() + + geom_smooth(method = "lm") + + ggtitle("Math Score vs Number of
Awards") + + theme_minimal() geom_smooth() using formula = 'y ~ x' > poisson_model <-
glm(num_awards ~ prog + math, + family = poisson(link = "log"), + data = data) > # Model
summary > summary(poisson_model)

Call: glm(formula = num_awards ~ prog + math, family = poisson(link = "log"), data = data)

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.24712 0.65845 -7.969 1.60e-15 progAcademic 1.08386 0.35825 3.025 0.00248
progVocational 0.36981 0.44107 0.838 0.40179
math 0.07015 0.01060 6.619 3.63e-11 — Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 287.67 on 199 degrees of freedom

Residual deviance: 189.45 on 196 degrees of freedom AIC: 373.5

Number of Fisher Scoring iterations: 6

## 23 Exponentiate coefficients

```
exp(coef(poisson_model)) (Intercept) progAcademic progVocational math
0.00526263 2.95606545 1.44745846 1.07267164 # Confidence Intervals
exp(confint(poisson_model)) Waiting for profiling to be done... 2.5 % 97.5
% (Intercept) 0.001400301 0.0185869 progAcademic 1.545030949 6.3972554
progVocational 0.612501440 3.5467192 math 1.050742170 1.0953533 # Predicted
number of awards datapredictedawards < -predict(poissonmodel, +type =
"response")head(data)idnumawardsprogmathpredictedawards1450Vocational410.1351909721080General4
[1] 189.4496 # Degrees of Freedom poisson_model$df.residual[1]196CheckOverdispersiondispersion <
-poissonmodeldeviance / + poisson_model$df.residual dispersion [1] 0.9665797 gg-
plot(data, + aes(x = math, + y = predicted_awards, + color = prog)) + +
geom_line() + + ggtitle("Predicted Number of Awards") + + ylab("Predicted Awards")
+ + theme_minimal()
```

## 24 Conclusion

Key Findings:

1. Math score is positively associated
2. Program type significantly affects
3. Students in academic programs tend
4. Students in academic programs tend to earn more awards compared to
5. Poisson regression is appropriate because Strong academic preparation and performance Educational Implication: Strong academic preparation and performance in mathematics may increase student achievement Educational Implication: Strong academic preparation and performance in mathematics may increase student achievement and recognition through awards. Educational Implication: Strong academic preparation and performance in mathematics may increase student achievement and recognition through awards.

Interpretation:

Overdispersion: The Negative Binomial is used because the variance of absences is much larger than the mean.

Findings: Higher math scores correlate with fewer absences. The Academic track has the best attendance records.

problem 7: School administrators study the attendance behavior (number of days absent) as a function of program type (e.g., vocational, general, or academic) and standardized math scores. The data set is taken from [https://stats.idre.ucla.edu/stat/data/nb\\_data.dta](https://stats.idre.ucla.edu/stat/data/nb_data.dta) . Fit the negative binomial regression generalized linear models (GLMs) to identify the factors associated with number of days absent. Interpret the result.

```
[27]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Load data
url = "https://stats.idre.ucla.edu/stat/data/nb_data.dta"

df = pd.read_stata(url)

# Convert prog to string
df["prog"] = df["prog"].astype(str)

# Fit model
model = smf.glm(
    formula="daysabs ~ math + prog",
    data=df,
    family=sm.families.NegativeBinomial()
).fit()

# Summary
print(model.summary())

# IRR
irr = np.exp(model.params)

print("\nIncident Rate Ratios (IRR):")
print(irr)
```

#### Generalized Linear Model Regression Results

```
=====
Dep. Variable:          daysabs    No. Observations:          314
Model:                  GLM        Df Residuals:              310
Model Family:          NegativeBinomial    Df Model:              3
Link Function:          Log           Scale:                  1.0000
Method:                 IRLS         Log-Likelihood:        -865.68
Date:                   Sat, 16 May 2026    Deviance:              350.98
Time:                   11:02:34          Pearson chi2:          331.
No. Iterations:         6                Pseudo R-squ. (CS):    0.1926
Covariance Type:        nonrobust
=====
```

|             | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|-------------|---------|---------|--------|-------|--------|--------|
| Intercept   | 2.6150  | 0.200   | 13.054 | 0.000 | 2.222  | 3.008  |
| prog[T.2.0] | -0.4408 | 0.185   | -2.379 | 0.017 | -0.804 | -0.078 |
| prog[T.3.0] | -1.2786 | 0.203   | -6.284 | 0.000 | -1.677 | -0.880 |
| math        | -0.0060 | 0.003   | -2.358 | 0.018 | -0.011 | -0.001 |

```
=====
```

Incident Rate Ratios (IRR):

```
Intercept      13.667374
prog[T.2.0]     0.643551
prog[T.3.0]     0.278418
math           0.994030
dtype: float64
```

C:\Users\Lenovo\anaconda3\Lib\site-

```
packages\statsmodels\genmod\family.py:1367: ValueWarning: Negative
binomial dispersion parameter alpha not set. Using default value alpha=1.0.
  warnings.warn("Negative binomial dispersion parameter alpha not "
```

## 25 In R

### 26 Load Libraries

```
library(ggplot2) library(haven) # Import .dta dataset data <- read_dta( +
"https://stats.idre.ucla.edu/stat/data/nb_data.dta" + ) ! curl package not installed,
falling back to using url() # View first rows head(data) # A tibble: 6 × 5 id gender
math daysabs prog <dbl+lbl> 1 1001 2 [male] 63 4 2 2 1002 2 [male] 27 4 2 3 1003 1
[female] 20 2 2 4 1004 1 [female] 16 3 2 5 1005 1 [female] 2 3 2 6 1006 1 [female] 71 13
2 # Structure str(data) tibble [314 × 5] (S3: tbl_df/tbl/data.frame) $ id : num [1:314]
1001 1002 1003 1004 1005 ... ..- attr(, "format.stata")= chr "%9.0g" $ gender : dbl+lbl
[1:314] 2, 2, 1, 1, 1, 1, 1, 2, 2, 2, 1, 2, 1, 1, 2, 2, 1, 1, 2, 2,... ..@ format.stata:
chr "%9.0g" ..@ labels : Named num [1:2] 1 2 .. ..- attr(, "names")= chr [1:2] "female"
"male" $ math : num [1:314] 63 27 20 16 2 71 63 3 51 49 ... ..- attr(, "label")= chr "ctbs
math pct rank" ..- attr(, "format.stata")= chr "%9.0g" $ daysabs: num [1:314] 4 4 2 3 3
13 11 7 10 9 ... ..- attr(, "label")= chr "number days absent" ..- attr(, "format.stata")=
chr "%9.0g" $ prog : num [1:314] 2 2 2 2 2 2 2 2 3 ... ..- attr(, "format.stata")= chr
"%9.0g" - attr(, "notes")= chr [1:2] "1" "Data on two LA high schools" # Convert program
into factor dataprog <- factor(dataprog, + levels = c(1,2,3), + labels = c("General",
+ "Academic", + "Vocational")) # Summary statistics summary(data) id gender math
daysabs prog
Min. :1001 Min. :1.00 Min. : 1.00 Min. : 0.000 General : 40
1st Qu.:1079 1st Qu.:1.00 1st Qu.:28.00 1st Qu.: 1.000 Academic :167
Median :1158 Median :1.00 Median :48.00 Median : 4.000 Vocational:107
Mean :1576 Mean :1.49 Mean :48.27 Mean : 5.955
3rd Qu.:2078 3rd Qu.:2.00 3rd Qu.:70.00 3rd Qu.: 8.000
Max. :2157 Max. :2.00 Max. :99.00 Max. :35.000
# Mean absences by program data %>% + group_by(prog) %>% + summarise( +
Mean_Absence = mean(daysabs), + SD_Absence = sd(daysabs), + Mean_Math =
mean(math) + ) # A tibble: 3 × 4 prog Mean_Absence SD_Absence Mean_Math
1 General 10.6 8.20 44.8 2 Academic 6.93 7.45 42.6 3 Vocational 2.67 3.73 58.4 #
Boxplot of absences by program ggplot(data, + aes(x = prog, + y = daysabs, + fill
= prog)) + + geom_boxplot() + + ggtitle("Days Absent by Program Type") + +
ylab("Days Absent") + + theme_minimal() # Scatter plot ggplot(data, + aes(x =
```

```

math, + y = daysabs)) + + geom_point() + + geom_smooth(method = "lm") +
+ ggtitle("Math Score vs Days Absent") + + theme_minimal() geom_smooth() using
formula = 'y ~ x' mean(datadaysabs)[1]5.955414var(datadaysabs) [1] 49.51877 pois-
son_model <- glm(daysabs ~ math + prog, + family = poisson, + data = data) sum-
mary(poisson_model)

```

Call: glm(formula = daysabs ~ math + prog, family = poisson, data = data)

Coefficients: Estimate Std. Error z value Pr(>|z|)

(Intercept) 2.651974 0.060736 43.664 < 2e-16 **math -0.006808 0.000931 -7.313 2.62e-13**  
progAcademic -0.439897 0.056672 -7.762 8.35e-15 **progVocational -1.281364 0.077886 -16.452**  
< **2e-16** — Signif. codes: 0 ‘’ **0.001** ‘’ 0.01 ‘’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 2217.7 on 313 degrees of freedom

Residual deviance: 1774.0 on 310 degrees of freedom AIC: 2665.3

Number of Fisher Scoring iterations: 5

```

nb_model <- glm.nb(daysabs ~ math + prog, + data = data) summary(nb_model)

```

Call: glm.nb(formula = daysabs ~ math + prog, data = data, init.theta = 1.032713156, link = log)

Coefficients: Estimate Std. Error z value Pr(>|z|)

(Intercept) 2.615265 0.197460 13.245 < 2e-16 **math -0.005993 0.002505 -2.392 0.0167**  
**progAcademic -0.440760 0.182610 -2.414 0.0158 \***  
**progVocational -1.278651 0.200720 -6.370 1.89e-10** — Signif. codes: 0 ‘’ **0.001** ‘’ 0.01 ‘\*’  
0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(1.0327) family taken to be 1)

Null deviance: 427.54 on 313 degrees of freedom

Residual deviance: 358.52 on 310 degrees of freedom AIC: 1741.3

Number of Fisher Scoring iterations: 1

```

      Theta: 1.033
    Std. Err.: 0.106

```

```

2 x log-likelihood: -1731.258 > exp(coef(nb_model)) (Intercept) math progAcademic
progVocational 13.6708448 0.9940249 0.6435471 0.2784127 > # Confidence Intervals >
exp(confint(nb_model)) Waiting for profiling to be done... 2.5 % 97.5 % (Intercept) 9.4126616
20.3470498 math 0.9891583 0.9989340 progAcademic 0.4448288 0.9115184 progVocational 0.1857247
0.4106239 > data$predicted_absence <- predict(nb_model, + type = "response") > head(data) #
A tibble: 6 x 6 id gender math daysabs prog predicted_absence <dbl>|lbl> 1 1001 2 [male] 63 4
Academic 6.03 2 1002 2 [male] 27 4 Academic 7.48 3 1003 1 [female] 20 2 Academic 7.80 4 1004 1
[female] 16 3 Academic 7.99 5 1005 1 [female] 2 3 Academic 8.69 6 1006 1 [female] 71 13 Academic
5.75 > ggplot(data, + aes(x = math, + y = predicted_absence, + color = prog)) + + geom_line()
+ + ggtitle("Predicted Days Absent") + + ylab("Predicted Absences") + + theme_minimal() >
AIC(poisson_model) [1] 2665.285 > AIC(nb_model) [1] 1741.258

```



## 27 Conclusion

Key Findings: 1. Math score is negatively associated 2. Program type significantly affects 3. Negative Binomial regression provides a better fit than Poisson regression Educational Implication: Students with lower academic performance Educational Implication: Students with lower academic performance may require attendance interventions Educational Implication: Students with lower academic performance may require attendance interventions and academic support programs. Educational Implication: Students with lower academic performance may require attendance interventions and academic support programs.

Interpretation:

Performance: Higher math scores correlate with fewer absences.

Program: Vocational and Academic students have significantly better attendance than those in the General track.

problem 8: The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, whether there were children in the group, and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish, so there are excess zeros in the data because of the people that did not fish. The data set is taken from <https://stats.idre.ucla.edu/stat/data/fish.csv>. Fit the Zero-Inflated Poisson regression generalized linear model (GLMs) to identify the factors associated with the number of fish caught. Interpret the result.

```
[18]: import pandas as pd
import statsmodels.api as sm

# 1. Load the data
url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
df = pd.read_csv(url)

# 2. Define variables
# Response: count (number of fish caught)
# Predictors for the Poisson part (count): child, camper, persons
# Predictors for the Logit part (zero-inflation): child, persons
y = df['count']
x_count = sm.add_constant(df[['child', 'camper', 'persons']])
x_zero = sm.add_constant(df[['child', 'persons']])

# 3. Fit the Zero-Inflated Poisson Model
model = sm.ZeroInflatedPoisson(y, x_count, exog_infl=x_zero, inflation='logit').
    ↪fit()

print(model.summary())
```

Optimization terminated successfully.

Current function value: 3.022141

Iterations: 23

Function evaluations: 25

Gradient evaluations: 25

#### ZeroInflatedPoisson Regression Results

```
=====
Dep. Variable:          count    No. Observations:          250
Model:          ZeroInflatedPoisson    Df Residuals:          246
Method:          MLE    Df Model:          3
Date:          Sat, 16 May 2026    Pseudo R-squ.:          0.3296
Time:          10:57:17    Log-Likelihood:          -755.54
converged:          True    LL-Null:          -1127.0
Covariance Type:          nonrobust    LLR p-value:          1.007e-160
=====
```

===

|                 | coef    | std err | z       | P> z  | [0.025 |
|-----------------|---------|---------|---------|-------|--------|
| 0.975]          |         |         |         |       |        |
| -----           |         |         |         |       |        |
| ---             |         |         |         |       |        |
| inflate_const   | 0.9813  | 0.414   | 2.370   | 0.018 | 0.170  |
| 1.793           |         |         |         |       |        |
| inflate_child   | 1.8227  | 0.315   | 5.780   | 0.000 | 1.205  |
| 2.441           |         |         |         |       |        |
| inflate_persons | -0.8454 | 0.190   | -4.456  | 0.000 | -1.217 |
| -0.474          |         |         |         |       |        |
| const           | -0.8452 | 0.166   | -5.103  | 0.000 | -1.170 |
| -0.521          |         |         |         |       |        |
| child           | -1.1472 | 0.091   | -12.539 | 0.000 | -1.327 |
| -0.968          |         |         |         |       |        |
| camper          | 0.7597  | 0.094   | 8.096   | 0.000 | 0.576  |
| 0.944           |         |         |         |       |        |
| persons         | 0.8337  | 0.043   | 19.587  | 0.000 | 0.750  |
| 0.917           |         |         |         |       |        |

=====

===

## 28 In R

## 29 Import dataset

```
fish <- read.csv( + "https://stats.idre.ucla.edu/stat/data/fish.csv" + ) # View first
rows head(fish) nofish livebait camper persons child xb zg count 1 1 0 0 1 0 -0.8963146
3.0504048 0 2 0 1 1 1 0 -0.5583450 1.7461489 0 3 0 1 0 1 0 -0.4017310 0.2799389 0 4 0 1
1 2 1 -0.9562981 -0.6015257 0 5 0 1 0 1 0 0.4368910 0.5277091 1 6 0 1 1 4 2 1.3944855
-0.7075348 0 # Structure of dataset str(fish) 'data.frame': 250 obs. of 8 variables: $
nofish : int 1 0 0 0 0 0 0 0 1 0 ... $ livebait: int 0 1 1 1 1 1 1 1 0 1 ... $ camper : int 0
1 0 1 0 1 0 0 1 1 ... $ persons : int 1 1 1 2 1 4 3 4 3 1 ... $ child : int 0 0 0 1 0 2 1 3 2
0 ... $ xb : num -0.896 -0.558 -0.402 -0.956 0.437 ... $ zg : num 3.05 1.746 0.28 -0.602
0.528 ... $ count : int 0 0 0 0 1 0 0 0 0 1 ... # Summary statistics summary(fish)
```

```

nofish livebait camper persons child
Min. :0.000 Min. :0.000 Min. :0.000 Min. :1.000 Min. :0.000
1st Qu.:0.000 1st Qu.:1.000 1st Qu.:0.000 1st Qu.:2.000 1st Qu.:0.000
Median :0.000 Median :1.000 Median :1.000 Median :2.000 Median :0.000
Mean :0.296 Mean :0.864 Mean :0.588 Mean :2.528 Mean :0.684
3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:4.000 3rd Qu.:1.000
Max. :1.000 Max. :1.000 Max. :1.000 Max. :4.000 Max. :3.000
xb zg count
Min. :-3.275050 Min. :-5.6259 Min. : 0.000
1st Qu.: 0.008267 1st Qu.: -1.2527 1st Qu.: 0.000
Median : 0.954550 Median : 0.6051 Median : 0.000
Mean : 0.973796 Mean : 0.2523 Mean : 3.296
3rd Qu.: 1.963855 3rd Qu.: 1.9932 3rd Qu.: 2.000
Max. : 5.352674 Max. : 4.2632 Max. :149.000
# Frequency of fish counts table(fish$count)

0 1 2 3 4 5 6 7 8 9 10 11 13 14 15 16 21 22 29 30 31 142 31 20 12 6
10 4 3 2 2 1 1 1 1 2 1 2 1 1 1 1 32 38 65 149 2 1 1 1 > # Proportion
of zero counts > mean(fish$count == 0)[1]0.568 > Histogramoffishcounts >
ggplot(fish, +aes(x = count)) + +geom_histogram(binwidth = 1, +fill = "skyblue", +color =
"black") + +ggtitle("DistributionofFishCounts") + +theme_minimal() > Numberofzeros >
sum(fish$count == 0) [1] 142 > poisson_model <- glm(count ~ child + persons + + camper
+ livebait, + family = poisson, + data = fish) > summary(poisson_model)

Call: glm(formula = count ~ child + persons + camper + livebait, family = poisson, data = fish)

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.39007 0.26490 -12.797 < 2e-16 child -1.70993 0.08140 -21.006 < 2e-16 persons
1.07396 0.03900 27.534 < 2e-16 camper 0.80009 0.08947 8.943 < 2e-16 livebait 1.66967
0.23318 7.160 8.05e-13 — Signif. codes: 0 ‘’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 2958.4 on 249 degrees of freedom

Residual deviance: 1246.9 on 245 degrees of freedom AIC: 1594

Number of Fisher Scoring iterations: 6

zip_model <- zeroinfl( + count ~ child + persons + + camper + livebait | + child +
persons, + data = fish, + dist = "poisson" + ) summary(zip_model)

Call: zeroinfl(formula = count ~ child + persons + camper + livebait | child + persons, data =
fish, dist = "poisson")

Pearson residuals: Min 1Q Median 3Q Max -2.52856 -0.71880 -0.40202 -0.06532 22.37738

Count model coefficients (poisson with log link): Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.40417 0.28374 -8.473 < 2e-16 child -1.18775 0.09377 -12.667 < 2e-16 persons
0.83764 0.04352 19.249 < 2e-16 camper 0.63160 0.09456 6.679 2.40e-11 livebait 1.73642
0.24366 7.126 1.03e-12 ***

Zero-inflation model coefficients (binomial with logit link): Estimate Std. Error z value Pr(>|z|)
(Intercept) 0.9096 0.4418 2.059 0.0395 *

```

child 1.9050 0.3394 5.612 2.00e-08 *persons* -0.8837 0.2037 -4.338 1.44e-05 — Signif. codes:  
0 ‘’ 0.001 ‘’ 0.01 ‘’ 0.05 ‘.’ 0.1 ‘’ 1

```
Number of iterations in BFGS optimization: 13 Log-likelihood: -715.3 on 8 Df >
AIC(poisson_model) [1] 1593.955 > AIC(zip_model) [1] 1446.531 > exp(coef(zip_model))
count_(Intercept) count_child count_persons count_camper 0.09034065 0.30490559 2.31091072
1.88061361 count_livebait zero_(Intercept) zero_child zero_persons 5.67698748 2.48343294
6.71941415 0.41326323 > fish$predicted_count <- predict(zip_model, + type = "response") >
head(fish) nofish livebait camper persons child xb zg count predicted_count 1 1 0 1 0 -0.8963146
3.0504048 0 0.1030292 2 0 1 1 1 0 -0.5583450 1.7461489 0 1.0999619 3 0 1 0 1 0 -0.4017310 0.2799389 0
0.5848952 4 0 1 1 2 1 -0.9562981 -0.6015257 0 0.4079221 5 0 1 0 1 0 0.4368910 0.5277091 1 0.5848952
6 0 1 1 4 2 1.3944855 -0.7075348 0 0.5987964 > # Conclusion Key Findings:
```

1. Fishing duration significantly increases
2. Excess zeros exist because many visitors
3. Zero-Inflated Poisson regression provides

Practical Implication: ZIP regression effectively separates: Practical Implication: ZIP regression effectively separates: - Visitors who never fish This improves modeling accuracy

Interpretation:

Interpretation:Non-Fishers: Groups with children are significantly more likely to be non-fishers.Catch Rate: Campers catch 2.14 times more fish than non-campers.

problem 9: The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, whether there were children in the group, and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish, so there are excess zeros in the data because of the people that did not fish. The data set is taken from <https://stats.idre.ucla.edu/stat/data/fish.csv>. Fit the Zero-Inflated Binomial regression generalized linear model (GLMs) to identify the factors associated with the number of awards earned by students at one high school. Interpret the result.

```
[19]: import pandas as pd
import statsmodels.api as sm

# 1. Load the dataset
url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
df = pd.read_csv(url)

# 2. Define the response and predictors
# count: Number of fish caught
# Predictors for count part: child, camper, persons
# Predictors for inflation part: child, persons
y = df['count']
x_count = sm.add_constant(df[['child', 'camper', 'persons']])
x_zero = sm.add_constant(df[['child', 'persons']])

# 3. Fit the Zero-Inflated Negative Binomial (ZINB) Model
```

```

# Note: ZINB is often preferred over ZIP if variance > mean
model = sm.ZeroInflatedPoisson(y, x_count, exog_infl=x_zero, inflation='logit').
    fit()
print(model.summary())

# Calculate Odds Ratios (OR) and Incident Rate Ratios (IRR)
params = model.params
print("\n--- Odds Ratios (Non-Fisher Probability) ---")
print(np.exp(params[params.index.str.contains('inflate')]))

print("\n--- Incident Rate Ratios (Catch Rate) ---")
print(np.exp(params[~params.index.str.contains('inflate')]))

```

Optimization terminated successfully.

Current function value: 3.022141

Iterations: 23

Function evaluations: 25

Gradient evaluations: 25

#### ZeroInflatedPoisson Regression Results

```

=====
Dep. Variable:          count    No. Observations:          250
Model:                ZeroInflatedPoisson    Df Residuals:          246
Method:                  MLE    Df Model:                  3
Date:                  Sat, 16 May 2026    Pseudo R-squ.:          0.3296
Time:                  10:57:25    Log-Likelihood:          -755.54
converged:                  True    LL-Null:                  -1127.0
Covariance Type:          nonrobust    LLR p-value:              1.007e-160
=====

```

```

=====
              coef      std err          z      P>|z|      [0.025
0.975]
-----
---
inflate_const      0.9813      0.414      2.370      0.018      0.170
1.793
inflate_child      1.8227      0.315      5.780      0.000      1.205
2.441
inflate_persons    -0.8454      0.190     -4.456      0.000     -1.217
-0.474
const              -0.8452      0.166     -5.103      0.000     -1.170
-0.521
child              -1.1472      0.091    -12.539      0.000     -1.327
-0.968
camper              0.7597      0.094      8.096      0.000      0.576
0.944
persons            0.8337      0.043     19.587      0.000      0.750
0.917
=====

```

===

--- Odds Ratios (Non-Fisher Probability) ---

```
inflate_const      2.667858
inflate_child      6.188700
inflate_persons    0.429402
dtype: float64
```

--- Incident Rate Ratios (Catch Rate) ---

```
const      0.429457
child      0.317512
camper     2.137712
persons    2.301896
dtype: float64
```

## 30 In R

```
library(psc1) library(MASS) library(dplyr) library(ggplot2) fish <- read.csv( +
"https://stats.idre.ucla.edu/stat/data/fish.csv" + ) # View first rows head(fish) nofish
livebait camper persons child xb zg count 1 1 0 0 1 0 -0.8963146 3.0504048 0 2 0 1 1
1 0 -0.5583450 1.7461489 0 3 0 1 0 1 0 -0.4017310 0.2799389 0 4 0 1 1 2 1 -0.9562981
-0.6015257 0 5 0 1 0 1 0 0.4368910 0.5277091 1 6 0 1 1 4 2 1.3944855 -0.7075348 0 #
Structure str(fish) 'data.frame': 250 obs. of 8 variables: $ nofish : int 1 0 0 0 0 0 0 1 0
... $ livebait: int 0 1 1 1 1 1 1 0 1 ... $ camper : int 0 1 0 1 0 1 0 0 1 1 ... $ persons
: int 1 1 1 2 1 4 3 4 3 1 ... $ child : int 0 0 0 1 0 2 1 3 2 0 ... $ xb : num -0.896 -0.558
-0.402 -0.956 0.437 ... $ zg : num 3.05 1.746 0.28 -0.602 0.528 ... $ count : int 0 0 0 0
1 0 0 0 0 1 ... summary(fish) nofish livebait camper persons child
Min. :0.000 Min. :0.000 Min. :0.000 Min. :1.000 Min. :0.000
1st Qu.:0.000 1st Qu.:1.000 1st Qu.:0.000 1st Qu.:2.000 1st Qu.:0.000
Median :0.000 Median :1.000 Median :1.000 Median :2.000 Median :0.000
Mean :0.296 Mean :0.864 Mean :0.588 Mean :2.528 Mean :0.684
3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:4.000 3rd Qu.:1.000
Max. :1.000 Max. :1.000 Max. :1.000 Max. :4.000 Max. :3.000
xb zg count
Min. :-3.275050 Min. :-5.6259 Min. : 0.000
1st Qu.: 0.008267 1st Qu.: -1.2527 1st Qu.: 0.000
Median : 0.954550 Median : 0.6051 Median : 0.000
Mean : 0.973796 Mean : 0.2523 Mean : 3.296
3rd Qu.: 1.963855 3rd Qu.: 1.9932 3rd Qu.: 2.000
Max. : 5.352674 Max. : 4.2632 Max. :149.000
# Frequency table table(fish$count)
0 1 2 3 4 5 6 7 8 9 10 11 13 14 15 16 21 22 29 30 31 142 31 20 12 6 10 4 3 2 2 1 1 1 1 2 1 2 1 1 1 1 32 38 65
149 2 1 1 1 > # Proportion of zeros > mean(fish$count == 0)[1]0.568 > Histogramof fishcounts >
ggplot(fish, +aes(x = count)) + +geom_histogram(binwidth = 1, +fill = "skyblue", +color =
"black") + +ggtitle("DistributionofFishCounts") + +theme_minimal() > mean(fish$count) [1]
3.296 > var(fish$count) [1] 135.3739 > poisson_model <- glm( + count ~ child + persons + +
camper + livebait, + family = poisson, + data = fish + ) > summary(poisson_model)
```

Call: glm(formula = count ~ child + persons + camper + livebait, family = poisson, data = fish)

Coefficients: Estimate Std. Error z value Pr(>|z|)

(Intercept) -3.39007 0.26490 -12.797 < 2e-16 **child -1.70993 0.08140 -21.006 < 2e-16** persons  
1.07396 0.03900 27.534 < 2e-16 **camper 0.80009 0.08947 8.943 < 2e-16** livebait 1.66967  
0.23318 7.160 8.05e-13 — **Signif. codes: 0 ‘’ 0.001 ‘\*\*’ 0.01 ‘\*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 2958.4 on 249 degrees of freedom

Residual deviance: 1246.9 on 245 degrees of freedom AIC: 1594

Number of Fisher Scoring iterations: 6

```
nb_model <- glm.nb( + count ~ child + persons + + camper + livebait, + data = fish  
+ ) summary(nb_model)
```

Call: glm.nb(formula = count ~ child + persons + camper + livebait, data = fish, init.theta =  
0.5155059824, link = log)

Coefficients: Estimate Std. Error z value Pr(>|z|)

(Intercept) -2.9760 0.4849 -6.138 8.37e-10 **child -1.8009 0.1839 -9.791 < 2e-16** persons 1.0618  
0.1108 9.581 < 2e-16 **camper 0.5179 0.2306 2.245 0.024752**  
**livebait 1.5379 0.4035 3.812 0.000138** — **Signif. codes: 0 ‘’ 0.001 ‘’ 0.01 ‘\*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

(Dispersion parameter for Negative Binomial(0.5155) family taken to be 1)

Null deviance: 423.64 on 249 degrees of freedom

Residual deviance: 209.88 on 245 degrees of freedom AIC: 809.16

Number of Fisher Scoring iterations: 1

```
Theta: 0.5155  
Std. Err.: 0.0811
```

```
2 x log-likelihood: -797.1610 > zinb_model <- zeroinfl( + count ~ child + persons + + camper +  
livebait | + child + persons, + dist = “negbin”, + data = fish + ) > summary(zinb_model)
```

Call: zeroinfl(formula = count ~ child + persons + camper + livebait | child + persons, data =  
fish, dist = “negbin”)

Pearson residuals: Min 1Q Median 3Q Max -0.77743 -0.56440 -0.32603 -0.05197 13.26347

Count model coefficients (negbin with log link): Estimate Std. Error z value Pr(>|z|)

(Intercept) -2.8627 0.4939 -5.796 6.79e-09 **child -1.1591 0.2664 -4.351 1.35e-05** persons  
1.0097 0.1202 8.397 < 2e-16 **camper 0.4791 0.2359 2.031 0.042273**  
**livebait 1.5278 0.3947 3.871 0.000109** \* Log(theta) -0.3969 0.2085 -1.904 0.056892 .

Zero-inflation model coefficients (binomial with logit link): Estimate Std. Error z value Pr(>|z|)

(Intercept) -2.0654 2.0828 -0.992 0.321374  
child 2.8463 0.8336 3.414 0.000639 \*\*\* persons -0.6063 0.5522 -1.098 0.272285  
— Signif. codes: 0 ‘’ **0.001** ‘’ 0.01 ‘’ 0.05 ‘.’ 0.1 ‘ ’ 1

Theta = 0.6724 Number of iterations in BFGS optimization: 18 Log-likelihood: -392.8 on 9 Df >  
AIC(poisson\_model) [1] 1593.955 > AIC(nb\_model) [1] 809.1614 > AIC(zinb\_model) [1] 803.6153

```
> exp(coef(zinb_model)) count_(Intercept) count_child count_persons count_camper 0.0571121
0.3137636 2.7449012 1.6146928 count_livebait zero_(Intercept) zero_child zero_persons 4.6080180
0.1267667 17.2240909 0.5453910 > fish$predicted_count <- predict( + zinb_model, + type =
"response" + ) > head(fish) nofish livebait camper persons child xb zg count predicted_count 1 1
0 0 1 0 -0.8963146 3.0504048 0 0.1466295 2 0 1 1 1 0 -0.5583450 1.7461489 0 1.0910016 3 0 1 0
1 0 -0.4017310 0.2799389 0 0.6756713 4 0 1 1 2 1 -0.9562981 -0.6015257 0 0.6090380 5 0 1 0 1 0
0.4368910 0.5277091 1 0.6756713 6 0 1 1 4 2 1.3944855 -0.7075348 0 0.5487991
```

## 31 Conclusion

Key Findings: 1. Fishing duration significantly affects fish-catching success. 2. Group composition influences fish counts. 3. Excess zeros are present because many visitors did not fish.

4. Overdispersion exists in the count data.
5. Zero-Inflated Negative Binomial model fits the data better than ordinary Poisson or Negative Binomial models. Practical Implication: ZINB regression effectively handles:

- Excess zero counts
- Overdispersed count data making it suitable for wildlife and recreational fishing studies.

Interpretation: The “Zero” Factor: The presence of children is the primary reason groups don’t fish at all. Groups with kids are over 6 times more likely to be “non-fishers.”

The “Catch” Factor: For those who actually fish, campers and larger groups catch significantly more fish. However, even if they fish, having children in the group reduces the expected catch by about 68%.

Bottom Line: To catch the most fish, go in a large group of adult campers.

Problem 10: A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from <https://stats.idre.ucla.edu/stat/data/ztp.dta> and fit the zero-truncated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

```
[20]: import pandas as pd
import pyreadstat
import requests

# Download file
url = "https://stats.oarc.ucla.edu/stat/data/ztp.dta"

r = requests.get(url)

with open("ztp.dta", "wb") as f:
    f.write(r.content)

# Read stata file
df, meta = pyreadstat.read_dta("ztp.dta")
```



```

print(df.head())

# 2. Fit the Negative Binomial GLM
# Predictors: age, hmo (insurance type), died (binary outcome)
# Response: stay (number of days)
model = smf.glm('stay ~ age + hmo + died', data=df,
                family=sm.families.NegativeBinomial()).fit()

# 3. Model Summary
print(model.summary())

# 4. Calculate Incident Rate Ratios (IRR)
# IRR = exp(coefficient)
irr = np.exp(model.params)
print("\n--- Incident Rate Ratios (IRR) ---")
print(irr)

```

```

      stay  age  hmo  died
0         4    4    0   0.0
1         9    4    1   0.0
2         3    7    1   1.0
3         9    6    0   0.0
4         1    7    0   1.0

```

#### Generalized Linear Model Regression Results

```

=====
Dep. Variable:          stay      No. Observations:          1493
Model:                  GLM      Df Residuals:              1489
Model Family:      NegativeBinomial  Df Model:                3
Link Function:          Log      Scale:                    1.0000
Method:              IRLS      Log-Likelihood:          -4955.3
Date:                Sat, 16 May 2026  Deviance:            827.73
Time:                10:57:29      Pearson chi2:          938.
No. Iterations:          5      Pseudo R-squ. (CS):      0.01154
Covariance Type:      nonrobust
=====

```

|           | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|-----------|---------|---------|--------|-------|--------|--------|
| Intercept | 2.4378  | 0.091   | 26.920 | 0.000 | 2.260  | 2.615  |
| age       | -0.0147 | 0.016   | -0.894 | 0.371 | -0.047 | 0.018  |
| hmo       | -0.1375 | 0.075   | -1.845 | 0.065 | -0.284 | 0.009  |
| died      | -0.2038 | 0.058   | -3.505 | 0.000 | -0.318 | -0.090 |

--- Incident Rate Ratios (IRR) ---

```

Intercept    11.447342
age           0.985360
hmo           0.871547

```

```
died          0.815615
dtype: float64

C:\Users\Lenovo\anaconda3\Lib\site-
packages\statsmodels\genmod\family.py:1367: ValueWarning: Negative
binomial dispersion parameter alpha not set. Using default value alpha=1.0.
  warnings.warn("Negative binomial dispersion parameter alpha not "
```

## 32 In R

## 33 Load Libraries

```
library(pscl) library(dplyr) library(ggplot2) library(haven) # Import .dta dataset data
<- read_dta( + "https://stats.idre.ucla.edu/stat/data/ztp.dta" + ) ! curl package not
installed, falling back to using url() # View first rows head(data) # A tibble: 6 × 4
stay age hmo died 1 4 4 0 0 2 9 4 1 0 3 3 7 1 1 4 9 6 0 0 5 1 7 0 1 6 4 5 0 1 # Structure of
dataset str(data) tibble [1,493 × 4] (S3: tbl_df/tbl/data.frame) $ stay: num [1:1493] 4
9 3 9 1 4 10 3 5 6 ... .. attr(, "label")= chr "Length of Stay" .. attr(, "format.stata")=
chr "%9.0g" $ age : num [1:1493] 4 4 7 6 7 5 8 7 4 4 ... .. attr(, "label")= chr "Age
Group" .. attr(, "format.stata")= chr "%9.0g" $ hmo : num [1:1493] 0 1 1 0 0 0 0 0 0 0
... .. attr(, "format.stata")= chr "%9.0g" $ died: num [1:1493] 0 0 1 0 1 1 1 1 0 0 ...
.. attr(, "format.stata")= chr "%9.0g" summary(data) stay age hmo died
Min. : 1.000 Min. :1.000 Min. :0.0000 Min. :0.0000
1st Qu.: 4.000 1st Qu.:4.000 1st Qu.:0.0000 1st Qu.:0.0000
Median : 8.000 Median :5.000 Median :0.0000 Median :0.0000
Mean : 9.729 Mean :5.234 Mean :0.1601 Mean :0.3429
3rd Qu.:13.000 3rd Qu.:6.000 3rd Qu.:0.0000 3rd Qu.:1.0000
Max. :74.000 Max. :9.000 Max. :1.0000 Max. :1.0000
# Check minimum stay min(data$stay)[1]1Frequencytabletable(data$stay)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 126 71 75 104 123 97 116 92 74 89 70 70 43 49 41 43
29 23 24 19 18 22 23 24 25 26 27 28 29 30 31 32 33 34 36 42 43 44 46 48 49 50 15 10 11 4 7 7 5 3 1 2 6 2 5
1 1 1 2 3 1 1 1 52 57 59 60 63 65 70 74 1 1 1 1 1 1 1 1 > # Distribution of hospital stay > ggplot(data,
+ aes(x = stay)) + + geom_histogram(binwidth = 1, + fill = "skyblue", + color = "black") + +
ggtitle("Distribution of Hospital Stay") + + theme_minimal() > # Age vs Stay > ggplot(data, +
aes(x = age, + y = stay)) + + geom_point(alpha = 0.5) + + geom_smooth(method = "lm") + +
ggtitle("Age vs Length of Hospital Stay") + + theme_minimal() geom_smooth() using formula = 'y
~ x' > datahmo <- factor(datahmo) > datadied <- factor(datadied) > poisson_model <- glm(
+ stay ~ age + hmo + died, + family = poisson, + data = data + ) > summary(poisson_model)

Call: glm(formula = stay ~ age + hmo + died, family = poisson, data = data)

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) 2.435763 0.027322 89.149 < 2e-16 age -0.014431 0.005033 -2.867 0.00414 hmo1
-0.135746 0.023722 -5.722 1.05e-08 died1 -0.203573 0.018360 -11.088 < 2e-16 — Signif.
codes: 0 ' ' 0.001 ' ' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 8296.7 on 1492 degrees of freedom
```

Residual deviance: 8115.7 on 1489 degrees of freedom AIC: 13826

Number of Fisher Scoring iterations: 5 # Step 5: Fit Zero-Truncated Poisson Model

```
ztp_model <- zerotrunc( stay ~ age + hmo + died, dist = "poisson", data = data )  
summary(ztp_model)
```

### **34 Step 6: Compare Models**

```
AIC(poisson_model)
```

```
AIC(ztp_model)
```

### **35 Lower AIC indicates better fit.**

### **36 Step 7: Incidence Rate Ratios (IRR)**

```
exp(coef(ztp_model))
```

### **37 Confidence Intervals**

```
exp(confint(ztp_model))
```

### **38 Step 8: Interpretation**

#age: #Positive coefficient indicates that older patients tend to stay longer in the hospital.

hmo: Insurance type influences hospital stay duration.

died: Patients who died during hospitalization may have significantly different lengths of stay.

### **39 Step 9: Predicted Length of Stay**

```
data$predicted_stay <- predict( ztp_model, type = "response" )  
head(data)
```

### **40 Step 10: Visualization of Predictions**

```
ggplot(data, aes(x = age, y = predicted_stay, color = died)) + geom_line() + ggtitle("Predicted  
Hospital Stay by Age") + ylab("Predicted Length of Stay") + theme_minimal()
```

### **41 Step 11: Goodness of Fit**

### **42 Residual Deviance**

```
ztp_model$deviance
```

## 43 Step 12: Conclusion

Key Findings: 1. Older patients generally have longer hospital stays.

2. Health insurance type significantly affects hospital stay duration.

3. Mortality status is associated with differences in stay length.

4. Zero-Truncated Poisson regression is appropriate because hospital stay cannot be zero days.

Practical Implication: Understanding factors affecting hospital stay can help improve healthcare planning, resource allocation, and patient management.

```
[21]: !pip install pyreadstat
```

Requirement already satisfied: pyreadstat in c:\users\lenovo\anaconda3\lib\site-packages (1.3.4)

Requirement already satisfied: narwhals>=2.10.1 in

c:\users\lenovo\anaconda3\lib\site-packages (from pyreadstat) (2.21.0)

Requirement already satisfied: numpy in c:\users\lenovo\anaconda3\lib\site-packages (from pyreadstat) (2.3.5)

Interpretation:

Drivers: Insurance (HMO) and Death are the key predictors.

Findings: Patients who died or had HMO insurance had significantly shorter hospital stays than survivors or privately insured patients.

Problem 11: A study of length of hospital stay, in days, as a function of age, kind of health insurance (HMO), and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from <https://stats.idre.ucla.edu/stat/data/ztp.dta>. Fit the zero-truncated negative binomial regression generalized linear models (GLMs) to identify the factors associated with length of hospital stay. Interpret the result.

```
[22]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import pyreadstat
import requests

from statsmodels.discrete.truncated_model import TruncatedLFNegativeBinomialP

# Download dataset
url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"

r = requests.get(url)

with open("ztp.dta", "wb") as f:
    f.write(r.content)
```

```

# Read data
df, meta = pyreadstat.read_dta("ztp.dta")

print(df.head())

# Response variable
y = df["stay"]

# Predictor variables
X = df[["age", "hmo", "died"]]

# Add intercept
X = sm.add_constant(X)

# Fit Zero-Truncated Negative Binomial model
model = TruncatedLFNegativeBinomialP(y, X)

result = model.fit()

# Show results
print(result.summary())

```

|   | stay | age | hmo | died |
|---|------|-----|-----|------|
| 0 | 4    | 4   | 0   | 0.0  |
| 1 | 9    | 4   | 1   | 0.0  |
| 2 | 3    | 7   | 1   | 1.0  |
| 3 | 9    | 6   | 0   | 0.0  |
| 4 | 1    | 7   | 0   | 1.0  |

Optimization terminated successfully.

Current function value: 3.185050

Iterations: 16

Function evaluations: 18

Gradient evaluations: 18

TruncatedLFNegativeBinomialP Regression Results

```

=====
=====
Dep. Variable:                stay    No. Observations:
1493
Model:                TruncatedLFNegativeBinomialP    Df Residuals:
1488
Method:                MLE    Df Model:
3
Date:                Sat, 16 May 2026    Pseudo R-squ.:
0.003263
Time:                10:57:33    Log-Likelihood:
-4755.3
converged:                True    LL-Null:
-4770.8

```

Covariance Type: nonrobust LLR p-value:  
7.955e-07

|       | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|-------|---------|---------|--------|-------|--------|--------|
| const | 2.4083  | 0.072   | 33.457 | 0.000 | 2.267  | 2.549  |
| age   | -0.0157 | 0.013   | -1.197 | 0.231 | -0.041 | 0.010  |
| hmo   | -0.1471 | 0.059   | -2.484 | 0.013 | -0.263 | -0.031 |
| died  | -0.2178 | 0.046   | -4.718 | 0.000 | -0.308 | -0.127 |
| alpha | 0.5663  | 0.031   | 18.132 | 0.000 | 0.505  | 0.628  |

## 44 In R

## 45 Load Libraries

```
library(pscI) library(MASS) library(dplyr) library(ggplot2) library(haven) # Import .dta dataset
> data <- read_dta( + "https://stats.idre.ucla.edu/stat/data/ztp.dta" + ) ! curl package not
installed, falling back to using url() > # View first rows > head(data) # A tibble: 6 × 4 stay
age hmo died 1 4 4 0 0 2 9 4 1 0 3 3 7 1 1 4 9 6 0 0 5 1 7 0 1 6 4 5 0 1 > # Structure of dataset
> str(data) tibble [1,493 × 4] (S3: tbl_df/tbl/data.frame) $ stay: num [1:1493] 4 9 3 9 1 4 10 3
5 6 ... .. attr(,"label")= chr "Length of Stay" .. attr(,"format.stata")= chr "%9.0g" $ age : num
[1:1493] 4 4 7 6 7 5 8 7 4 4 ... .. attr(,"label")= chr "Age Group" .. attr(,"format.stata")= chr
"%9.0g" $ hmo : num [1:1493] 0 1 1 0 0 0 0 0 0 0 ... .. attr(,"format.stata")= chr "%9.0g" $ died:
num [1:1493] 0 0 1 0 1 1 1 1 0 0 ... .. attr(,"format.stata")= chr "%9.0g" > summary(data) stay
age hmo died
Min. : 1.000 Min. :1.000 Min. :0.0000 Min. :0.0000
1st Qu.: 4.000 1st Qu.:4.000 1st Qu.:0.0000 1st Qu.:0.0000
Median : 8.000 Median :5.000 Median :0.0000 Median :0.0000
Mean : 9.729 Mean :5.234 Mean :0.1601 Mean :0.3429
3rd Qu.:13.000 3rd Qu.:6.000 3rd Qu.:0.0000 3rd Qu.:1.0000
Max. :74.000 Max. :9.000 Max. :1.0000 Max. :1.0000
> # Minimum stay > min(data$stay)[1]1 > Meanandvariance > mean(data$stay) [1]
9.728734 > var(data$stay)[1]66.14419 > Histogramofstaylength > ggplot(data,+aes(x =
stay)) + +geom_histogram(binwidth = 1,+fill = "skyblue",+color = "black") +
+ggtitle("DistributionofHospitalStay") + +theme_minimal() > AgevsStay >
ggplot(data,+aes(x = age,+y = stay)) + +geom_point(alpha = 0.5) + +geom_smooth(method =
"lm") + +ggtitle("AgevsHospitalStay") + +theme_minimal()+'geom_smooth()'using formula ='
y x' > datahmo <- factor(data$hmo) > datadied <- factor(data$died) > poisson_model <- glm( +
stay ~ age + hmo + died, + family = poisson, + data = data + ) > summary(poisson_model)

Call: glm(formula = stay ~ age + hmo + died, family = poisson, data = data)

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) 2.435763 0.027322 89.149 < 2e-16 age -0.014431 0.005033 -2.867 0.00414 hmo1
-0.135746 0.023722 -5.722 1.05e-08 died1 -0.203573 0.018360 -11.088 < 2e-16 — Signif.
codes: 0 ' 0.001 ' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 8296.7 on 1492 degrees of freedom

Residual deviance: 8115.7 on 1489 degrees of freedom AIC: 13826

Number of Fisher Scoring iterations: 5

```
nb_model <- glm.nb( + stay ~ age + hmo + died, + data = data + ) summary(nb_model)
```

Call: glm.nb(formula = stay ~ age + hmo + died, data = data, init.theta = 2.186744615, link = log)

Coefficients: Estimate Std. Error z value Pr(>|z|)

(Intercept) 2.43748 0.06449 37.797 < 2e-16 **age -0.01470 0.01176 -1.251 0.2111**

**hmo1 -0.13727 0.05335 -2.573 0.0101**

**died1 -0.20378 0.04159 -4.900 9.58e-07** — Signif. codes: 0 ‘’ 0.001 ‘’ 0.01 ‘\*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(2.1867) family taken to be 1)

Null deviance: 1602.5 on 1492 degrees of freedom

Residual deviance: 1568.4 on 1489 degrees of freedom AIC: 9614.9

Number of Fisher Scoring iterations: 1

```
Theta: 2.1867
Std. Err.: 0.0966
```

2 x log-likelihood: -9604.9240 > ztp\_model <- zerotrunc( + stay ~ age + hmo + died, + dist = “poisson”, + data = data + ) + # Step 6: Fit Zero-Truncated Poisson Model

```
ztp_model <- zerotrunc( stay ~ age + hmo + died, dist = “poisson”, data = data )
```

```
summary(ztp_model)
```

## 46 Step 7: Fit Zero-Truncated Negative Binomial Model

```
ztnb_model <- zerotrunc( stay ~ age + hmo + died, dist = “negbin”, data = data )
```

```
summary(ztnb_model)
```

## 47 Step 8: Model Comparison

```
AIC(poisson_model)
```

```
AIC(nb_model)
```

```
AIC(ztp_model)
```

```
AIC(ztnb_model)
```

**48 Lower AIC indicates better fit.**

**49 Step 9: Incidence Rate Ratios (IRR)**

```
exp(coef(ztnb_model))
```

**50 Confidence Intervals**

```
exp(confint(ztnb_model))
```

**51 Step 10: Interpretation**

age: Positive coefficient indicates that older patients tend to stay longer in the hospital.

hmo: Type of health insurance affects duration of hospital stay.

died: Patients who died during hospitalization may have significantly different lengths of stay.

**52 Step 11: Predicted Hospital Stay**

```
data$predicted_stay <- predict( ztnb_model, type = "response" )
```

```
head(data)
```

**53 Step 12: Visualization of Predictions**

```
ggplot(data, aes(x = age, y = predicted_stay, color = died)) + geom_line() + ggtitle("Predicted  
Hospital Stay by Age") + ylab("Predicted Stay Length") + theme_minimal()
```

**54 Step 13: Conclusion**

Key Findings: 1. Older age is associated with longer hospital stays.

2. Insurance type significantly affects length of hospitalization.

3. Mortality status influences hospital stay duration.

4. Overdispersion exists in the data, making Negative Binomial regression more appropriate than Poisson regression.

5. Since hospital stay cannot be zero, Zero-Truncated models are appropriate.

6. Zero-Truncated Negative Binomial (ZTNB) model provides the best fit due to:

- no zero counts
- overdispersion

Practical Implication: Understanding factors affecting hospital stay duration can improve hospital resource management, treatment planning, and patient care.



Interpretation: Significant Drivers: Mortality (Death) and HMO Insurance are the main factors that shorten hospital stays. Patients who pass away or have HMO coverage have stays that are 15% to 20% shorter than survivors or those with private insurance.

Non-Significant Factor: Age does not statistically impact the length of stay in this model.

Bottom Line: Hospital duration is driven by clinical outcomes and insurance type rather than the patient's age.

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